

Credit Risk Analysis and Prediction

Project Report

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StudyBuild

2026

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1. Narrative Analysis

1.1 Initial Data Cleaning

Number of records: exactly 30,000 observations, matching the official documentation.

Number of columns: 25 (24 input fields and 1 target variable).

Data types: all columns are stored as int64 integers. The original dataset contains no text or date fields.

Missing values: none. No fillna or dropna operation is required.

Duplicate rows: none. After using ID as the index, some customers have identical feature patterns, but they have different identifiers and represent different customers; therefore, they are not true duplicates.

Duplicate IDs: none. Each customer appears only once in the dataset.

Nonstandard category codes: EDUCATION codes 5 and 6 were mapped to code 0 rather than removed, and code 0 was treated as Unknown. The same consolidation approach was used for other anomalous categorical codes.

1.2 Distribution of Demographic Variables

Customer distribution by sex: 18,112 women and 11,888 men.

Customer distribution by education: 345 Unknown, 10,585 Graduate School, 14,030 University, 4,917 High School, and 123 Other.

Customer distribution by marital status: 54 Unknown, 13,659 Married, 15,964 Single, and 323 Other.

Customer distribution by default status: 23,364 customers did not default and 6,634 customers defaulted.

Demographic summary: most customers are women, single, and university educated. About 22% defaulted in the following month, which creates a meaningful class imbalance. The majority non-default class can dominate accuracy, so the modeling stage must emphasize metrics such as precision and recall and consider training-only resampling methods.

1.3 Descriptive Statistics

Credit limit: the distribution is clearly right-skewed; most customers have lower limits, while a small group has very high limits. Excess kurtosis is relatively low at approximately 0.54.

Payment status: the negative means indicate that most customers paid on time or early. However, the distributions are skewed because a smaller group experienced severe payment delays.

Bill and payment amounts: skewness is very high for some variables, with values such as 14.67 and 2.66. Most customers make relatively small payments, while a few make very large payments. The extremely high kurtosis also indicates strong tail outliers; approximately 75% of payments are below 5,000, but some values are much higher.

Age: the distribution is concentrated and comparatively regular, with most customers between approximately 28 and 41 years old.

Descriptive-statistics summary: the financial variables, especially bill and payment amounts, are highly skewed and contain many outliers. This supports the use of scaling and outlier-resistant preprocessing such as RobustScaler during modeling.

1.4 Payment Behavior Analysis

Among non-defaulting customers, the largest PAY_0 category is PAY_0 = 0 at 54.99%. Among defaulting customers, PAY_0 = 0 accounts for 28.45%, closely followed by PAY_0 = 2 at 27.79%.

The cross-tabulation shows that 51.67% of defaulting customers were delayed in the current month, compared with 14.51% of non-defaulting customers. Current-month delay is therefore about 3.6 times more common in the default group. The gap becomes smaller for older months, showing that recent delays are more useful for identifying high-risk customers.

Payment-behavior summary: customers with a two-month current delay (PAY_0 = 2) make up 27.79% of the default group but only 3.52% of the non-default group. Their relative presence in the default group is therefore about eight times higher, confirming the strong discriminatory value of PAY_0.

The separability analysis shows current-month delay rates of 51.67% for defaulting customers and 14.51% for non-defaulting customers, a gap of 37.17 percentage points. This gap gradually decreases for older months. The most recent payment-status variables, especially PAY_0, should therefore receive priority in an early-warning system.

1.5 Demographic Group Analysis

Sex: women have an average credit limit of 170,860 and a delay rate of 23.5%; men have an average credit limit of 163,520 and a delay rate of 35.2%. In this dataset, women receive slightly higher limits and show more timely payment behavior.

Marital status: married customers have the highest average credit limit (182,201) and a 23.5% default rate. Single customers have an average limit of 156,410 and a default rate of about 20.9%. The Other group has the lowest average limit (98,080), the highest delay rate (36.8%), and a 26% default rate.

Age: the youngest group (20–27) has the lowest average credit limit (100,824), the highest delay rate (38.3%), and a 24.2% default rate. Customers aged 31–37 have the highest average limit (198,438), the lowest delay rate (31.1%), and a 20.5% default rate.

High-Risk Groups (High Default Rate)

Married men aged 20–27 with a high-school education: 52% default rate, average credit limit of 85,600, and median limit of 50,000.

Single men aged 20–27 with a university education: 28.6% default rate, average credit limit of 72,945, and median limit of 50,000.

Single men aged 20–27 with a high-school education: 28.2% default rate, average credit limit of 59,315, and median limit of 50,000.

Married women aged 20–27 with a high-school education: 28.2% default rate, average credit limit of 84,426, and median limit of 55,000.

Medium-Risk Groups (Moderate Default Rate)

Married women aged 31–37 with a university education: 20.7% default rate, average credit limit of 173,292, and median limit of 150,000.

Married men aged 43–79 with a graduate education: 23.4% default rate, average credit limit of approximately 271,980, and median limit of 250,000.

Low-Risk Groups (Low Default Rate)

Married women aged 31–37 with a graduate education: 18.8% default rate, average credit limit of 250,572, and median limit of 230,000.

Married women aged 37–43 with a graduate education: 18.7% default rate, average credit limit of 248,163, and median limit of 230,000.

Single women aged 31–37 with a graduate education: 17.2% default rate, average credit limit of 246,048, and median limit of 225,000.

Single women aged 31–37 with a graduate education are the lowest-risk demographic segment, while married men aged 20–27 with a high-school education are the highest-risk segment.

Credit and risk gap: based on median credit limits, high-risk groups generally receive 50,000–55,000, whereas low-risk groups receive 225,000–230,000. This four- to five-fold gap suggests that the credit-allocation process already recognizes some risk, but high-risk groups still show high default rates despite their lower limits and require stronger controls.

Other influential factors: beyond demographics, payment behavior shows that current payment status (PAY_0) is the strongest predictor. Customers delayed in the current month are about 3.6 times more common in the default group, and a two-month delay increases relative presence by up to eight times. The composite repayment_ratio has a median of 0.095, meaning half of customers repay less than 10% of their debt balance.

Recommended Credit Policy

- Apply lower credit limits, risk-based pricing within applicable regulations, and closer monitoring to high-risk groups.
- Offer higher credit limits and more favorable terms to verified low-risk groups, subject to policy and fairness controls.
- Use models that combine approved customer attributes with payment behavior to improve risk detection, while monitoring fairness and regulatory compliance.

Demographic-group summary: current payment status (PAY_0) is the most important predictor of default. In this sample, the highest-risk demographic profiles are mainly young men aged 20–27 with high-school or university education, with default rates between 28% and 52%. The lowest-risk profiles are women aged 31–43 with graduate education, with default rates of approximately 17%–19%. Median credit limits differ by a factor of four to five between these groups.

For high-risk customers, apply stricter credit limits, shorter repayment periods, appropriate collateral or guarantees, secured or prepaid credit products, and continuous monitoring. Any pricing or eligibility decision should follow applicable regulations and fairness requirements.

For medium-risk customers, retain standard terms, perform periodic monitoring, and use an early-warning process for initial payment delays.

For low-risk customers, consider higher credit limits, lower rates, longer repayment periods, and loyalty benefits, subject to standard approval rules.

Financial education and debt-management support should also be offered to customers, especially those at higher risk. Combining these measures with machine-learning models can improve prediction and reduce credit losses.

Important: install PySide6 before running the code because the Matplotlib backend is set to QtAgg.

2. Answers to the Risk Manager's 10 Questions

This section summarizes the main project findings. Logistic regression is the primary default-prediction model; clustering, UMAP, GMM, and Random Forest provide supplementary behavioral analysis.

2.1 Question 1: What Is the Overall Condition of the Data?

The dataset contains 30,000 customers and 23 predictor variables. It has no missing values, duplicate IDs, or fully duplicated rows. The default rate is 22.12%, so the target is imbalanced. Financial amounts are skewed and contain outliers. Special EDUCATION, MARRIAGE, and PAY codes were retained where meaningful or consolidated into an Unknown category during cleaning.

2.2 Question 2: How Do Defaulting and Non-Defaulting Customers Differ?

Among defaulting customers, 64.83% have a history of payment delay, compared with 24.68% of non-defaulting customers. The average number of delayed months is about 2.0 versus 0.5. Six-month payments and credit limits are lower in the default group, while bill amount alone provides weaker separation. Exploratory tests with GMM, t-SNE, and K-means supported a useful three-segment solution, discussed later in the report.

2.3 Question 3: What Is the Most Important Repayment Behavior?

PAY_0 is the strongest warning signal: the current-delay rate is 51.67% among defaulting customers and 14.51% among non-defaulting customers. PAY_2 through PAY_6 follow, with weaker separation for older months. The composite repayment indicator and PCA were used only for exploratory analysis; the final model was trained on the original 23 predictors.

2.4 Question 4: How Do the Baseline and Optimized Models Perform?

The data was split into 24,000 training observations and 6,000 test observations. Continuous variables were processed with winsorization, Yeo–Johnson transformation, and RobustScaler; nominal variables

were one-hot encoded. SMOTENC was applied only within the training data and each cross-validation training fold. Optuna tuned the logistic-regression C parameter over 20 trials and selected 0.0114938.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline Logistic Regression	0.8033	0.6450	0.2464	0.3566	0.7355
Logistic Regression with SMOTENC	0.7493	0.4506	0.6081	0.5176	0.7590

Result: the SMOTENC model has lower overall accuracy, but it clearly improves recall and F1 for the default class. It is therefore more suitable when detecting high-risk customers is the main objective.

2.5 Question 5: Which Evaluation Metrics Are Appropriate?

Accuracy alone is not sufficient because approximately 78% of the data belongs to the non-default class. A model can appear accurate while performing poorly on the smaller default class.

Precision measures how many risk alerts are correct; recall measures the share of defaults detected; F1 balances precision and recall; ROC-AUC measures overall ranking ability; and PR-AUC focuses on performance for the default class. Cross-validation was performed with StratifiedKFold to preserve the class ratio across folds and estimate performance more reliably.

A standalone Random Forest default model was also tested before clustering, but it did not provide a meaningful improvement and was removed from the final default-modeling pipeline.

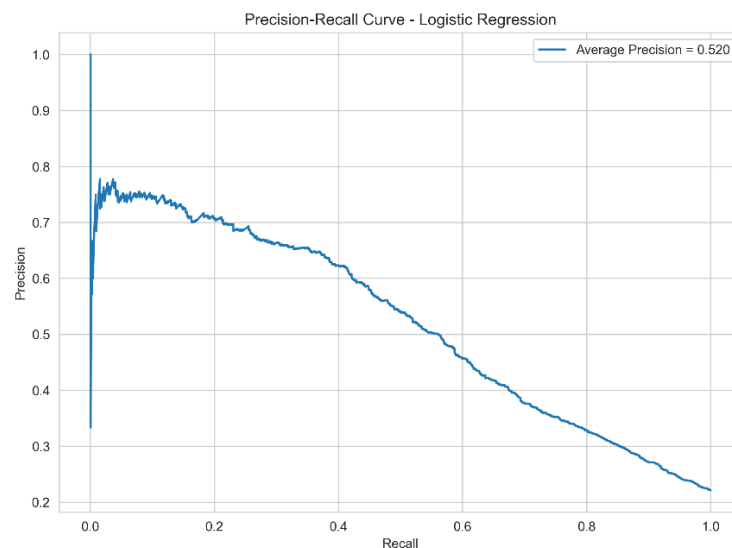


Figure 1

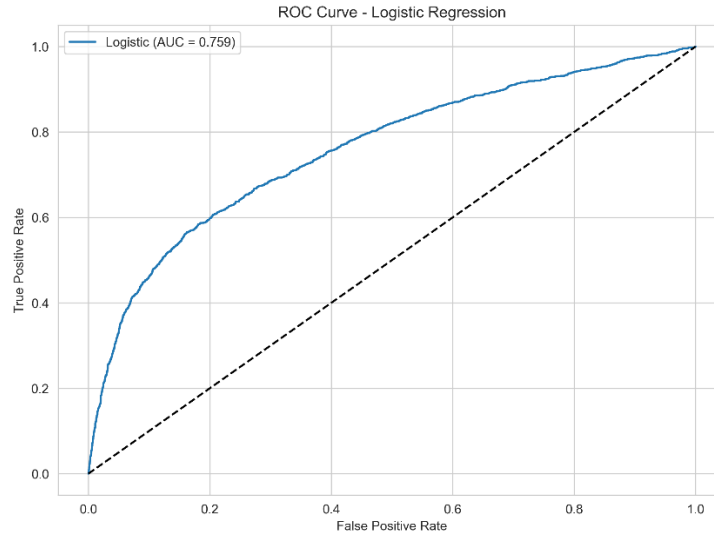


Figure 2

2.6 Question 6: Which Errors Matter Most to the Business?

A false negative occurs when a customer who will default is not detected and usually creates the larger credit loss. A false positive occurs when a healthy customer is incorrectly flagged as high risk, creating review costs or customer dissatisfaction. At a 0.50 threshold, the baseline model misses 1,000 defaults, while the SMOTENC model reduces this number to 520 at the cost of more false alerts.

2.7 Question 7: How Are Customers Grouped by Risk?

The primary risk groups were created from probabilities produced by the logistic model without synthetic sampling. Cutoffs of 0.166717 and 0.325343 were selected using validation data so that the test set did not influence threshold selection.

Risk Group	Test Count	Mean Probability	Actual Default Rate	Total Count
Low Risk	2,920	10.40%	11.71%	14,594
Medium Risk	1,895	22.38%	18.42%	9,359
High Risk	1,185	50.79%	53.67%	6,047

The consistent increase in actual default rate from the low-risk group to the high-risk group shows that the segmentation is useful for prioritizing customers.

2.8 Question 8: Which Variables Have the Greatest Influence?

In the logistic model, which predicts default directly, $PAY_0 = 2$ had the largest positive coefficient (1.1902) and was the strongest sign of higher risk. $PAY_0 = 0$, credit limit, and the second payment amount had negative coefficients and were associated with lower predicted risk, while $BILL_AMT6$ had a positive coefficient. SHAP highlighted $PAY_0 = 0$, university education, $BILL_AMT6$, PAY_AMT2 , and $LIMIT_BAL$. The top ten RFE positions also included $PAY_0 = 2$, $PAY_4 = 2$, several PAY_0 states, $LIMIT_BAL$, $PAY_5 = 2$, PAY_AMT2 , and $BILL_AMT3$.

In the supplementary cluster analysis, Random Forest was trained to predict the labels generated by GMM. For the high-risk cluster, SHAP ranked PAY_3, PAY_2, PAY_0, PAY_4, PAY_6, and PAY_5 as the most important variables, followed by PAY_AMT2 and PAY_AMT1. Directional effects showed that higher PAY status values increased the probability of high-risk-cluster membership. Random Forest RFE ranked PAY_3, PAY_2, PAY_4, PAY_6, PAY_0, PAY_5, BILL_AMT5, PAY_AMT2, BILL_AMT2, and PAY_AMT1 in the top ten. Payment-delay history was therefore the main driver of high-risk-cluster identification.

Random Forest hyperparameters were tuned with 20 Optuna trials. Optuna selected 218 trees, a maximum depth of 15, at least 6 samples to split a node, and at least 1 sample per leaf. Because the code uses a fixed seed, an exact rerun will usually reproduce the same result. Changing the seed, number of trials, data, or library versions may select different parameters and slightly change some SHAP or RFE positions. The overall pattern of importance is therefore more meaningful than a one- or two-position difference. These are predictive associations: logistic-regression results refer to default probability, while Random Forest results refer to high-risk-cluster membership. Neither result alone establishes causality.

2.9 Question 9: How Sensitive Are the Results to the Decision Threshold?

Model	Threshold	Precision	Recall	F1
Baseline	0.3	0.5197	0.5072	0.5133
Baseline	0.5	0.6450	0.2464	0.3566
Baseline	0.7	0.7424	0.0739	0.1343
SMOTENC	0.3	0.2802	0.8862	0.4258
SMOTENC	0.5	0.4506	0.6081	0.5176
SMOTENC	0.7	0.6223	0.4084	0.4932

A 0.30 threshold detects more defaults but generates more false alerts. A 0.70 threshold is more conservative and misses many defaults. Among these three options, 0.50 provides a more practical balance for the SMOTENC model. The final threshold should be aligned with error costs and the organization's review capacity.

2.10 Question 10: What Management Actions Are Recommended?

The model should be used as a decision-support tool, not as an automatic system for accepting or rejecting credit applications. Its main purpose is to rank customers by risk and help the institution decide which cases require more attention. The final decision should combine the model's prediction with human review, the customer's financial situation, internal credit policies, and relevant banking regulations.

Customers classified as high risk should be prioritized for detailed human review. These customers may require closer payment monitoring, contact before payment due dates, and immediate follow-up after the first sign of delay. Depending on the customer's situation and the institution's policies, possible actions may include setting a lower initial credit limit, requesting an appropriate guarantee or collateral, shortening the review period, or offering a suitable repayment plan. The purpose of these actions is to reduce the probability of default while avoiding an automatic rejection based only on the model's output.

Medium-risk customers should remain under periodic monitoring. Their payment behavior, credit utilization, bill amounts, and recent repayment delays should be reviewed regularly. An early-warning system can be used to identify negative changes, such as an increase in delayed payments or a reduction in repayment amounts. If the customer's financial behavior improves, the risk level may be reduced. If the behavior becomes worse, the customer can be moved to a higher monitoring level.

Low-risk customers can usually follow the standard or faster credit process. However, being classified as low risk does not mean that monitoring should completely stop. Credit limits should still be based on repayment capacity, income, and previous payment history. Customers with stable repayment behavior may receive suitable benefits, such as better service conditions, gradual increases in credit limits, or non-cash loyalty rewards, provided that these actions follow Central Bank regulations and the institution's internal policies.

The predicted probability from Logistic Regression can be used as the main risk score. The final classification threshold should be selected according to business objectives and the cost of prediction errors. A lower threshold identifies more customers who may default, but it also sends more safe customers for unnecessary review. A higher threshold reduces the number of false alarms, but it may fail to detect some customers who will actually default. Therefore, the selected threshold should reflect the cost of missed defaults, the institution's operational capacity, and the number of cases that can be manually reviewed.

The clustering results can be used as an additional source of information. Clustering does not replace the default prediction model. Instead, it helps identify different behavioral profiles among customers. For example, two customers may have similar predicted default probabilities but different payment histories, bill amounts, credit limits, or repayment patterns. These customers may need different management actions. One group may require early reminders, while another may need a revised repayment plan or closer credit-limit control. This makes customer management more targeted and avoids treating all risky customers in exactly the same way.

Before operational deployment, the model should be tested on data from a different time period or another customer sample. Probability calibration should also be checked to make sure that the predicted risk values represent the actual default rates. Model performance, fairness across customer groups, data drift, and changes in important variables should be monitored regularly. The institution should also define a clear schedule for model review and retraining.

Sensitive demographic variables should not be used as the only reason for approving or rejecting a customer. All decisions must follow banking regulations, data-protection requirements, and internal governance rules. Human oversight should remain part of the final decision process, especially for high-risk or borderline cases.

Overall, the recommended strategy is to combine risk probabilities, appropriate decision thresholds, behavioral clustering, and human judgment. This approach can reduce missed defaults and financial losses while also preventing unnecessary restrictions on customers with acceptable repayment capacity.

3. Supplementary Clustering Analysis

Clustering does not replace the default model; its purpose is to identify behavioral customer profiles. The target variable was excluded from cluster formation. Training-set default rates were examined only after clustering to interpret and name the groups.

3.1 UMAP and GMM Method

After the same preprocessing, 31 transformed features—excluding the target—were passed to UMAP and reduced to three dimensions. UMAP used 44 neighbors and a minimum distance of 0.002757. A three-component GMM with full covariance was then fitted in the UMAP space. The Gaussian assumption concerns components in this three-dimensional representation; it does not require every original feature to be normally distributed. Yeo–Johnson only reduces skewness in continuous features.

The Shapiro–Wilk test still rejected univariate normality for all features after the Yeo–Johnson transformation. With a very large sample, this test is highly sensitive and can reject even small departures from normality. This does not invalidate the GMM analysis because GMM was fitted to the UMAP representation, and its assumption concerns the Gaussian components in that latent space rather than perfect bell-shaped distributions for every original variable.

3.2 Clustering Results

Cluster	Count	Training Default Rate	Test Default Rate	Main Characteristic
Low Risk — Cluster 0	11,788	10.32%	11.25%	More Regular Payments
Medium Risk — Cluster 2	10,326	16.06%	16.72%	Higher Credit Limit
High Risk — Cluster 1	7,886	47.51%	46.12%	More Delays and Lower Payments

The silhouette score is 0.4926 on the training set and 0.4940 on the test set. The close values and stable cluster default rates suggest that the segmentation is not limited to the training data, although cluster separation is moderate.

3.3 Role of Random Forest and Optuna

Random Forest was trained to reproduce the GMM labels, not to predict default directly. Across 20 trials, Optuna selected 218 trees, a maximum depth of 15, a minimum of 6 samples per split, and a minimum of 1 sample per leaf. Test accuracy is 0.9845 and mean cross-validation accuracy is 0.9892. RFE and SHAP show that PAY status variables are the main drivers of high-risk-cluster membership.

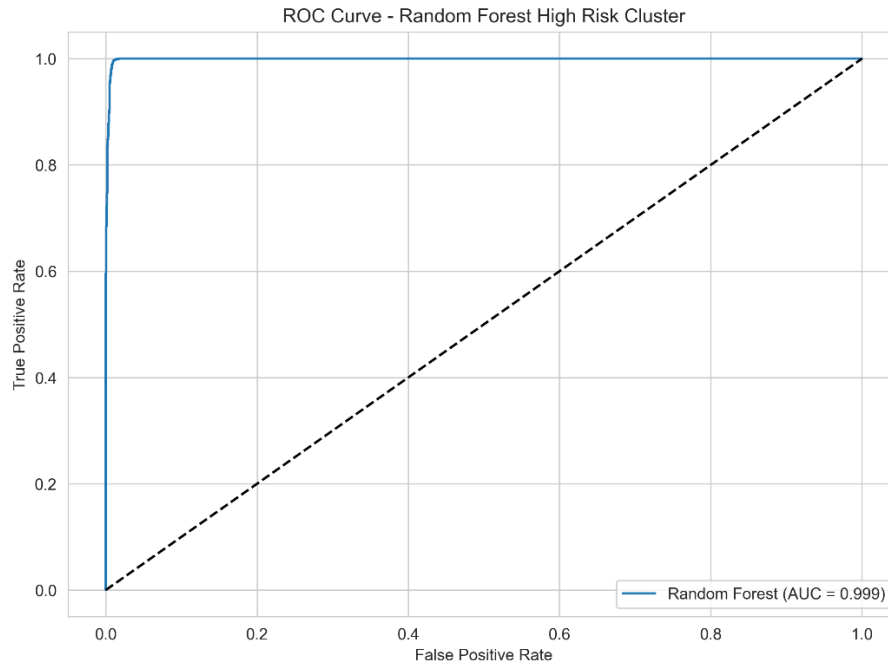


Figure 3

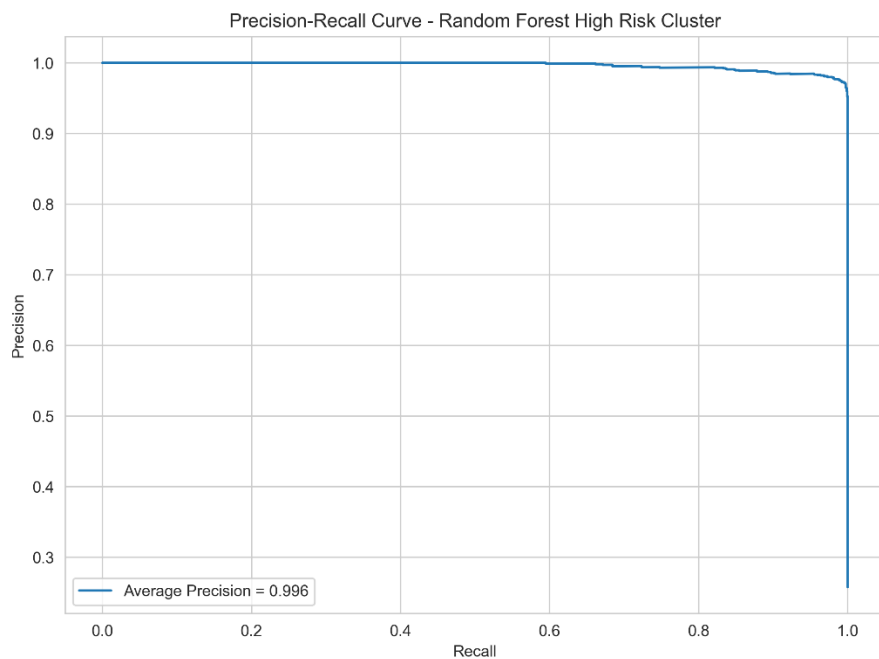


Figure 4

3.4 Clustering Visualizations and Summary

PCA, UMAP, and t-SNE visualizations were created for all 30,000 customers. The target variable was used only to color the points and was not included in dimensionality reduction. Optuna search for UMAP and GMM was disabled in the current run, and those parameters came from an earlier experiment. Logistic-

model probabilities should remain the primary basis for risk ranking; clustering and Random Forest are supplementary tools for behavioral segmentation.

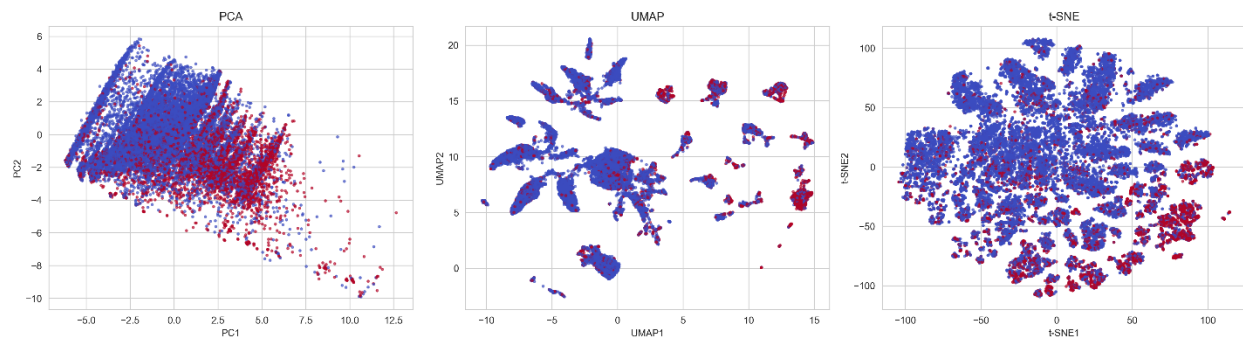


Figure 5

Figure 5 shows that UMAP provides clearer visual separation than PCA and t-SNE for this dataset.

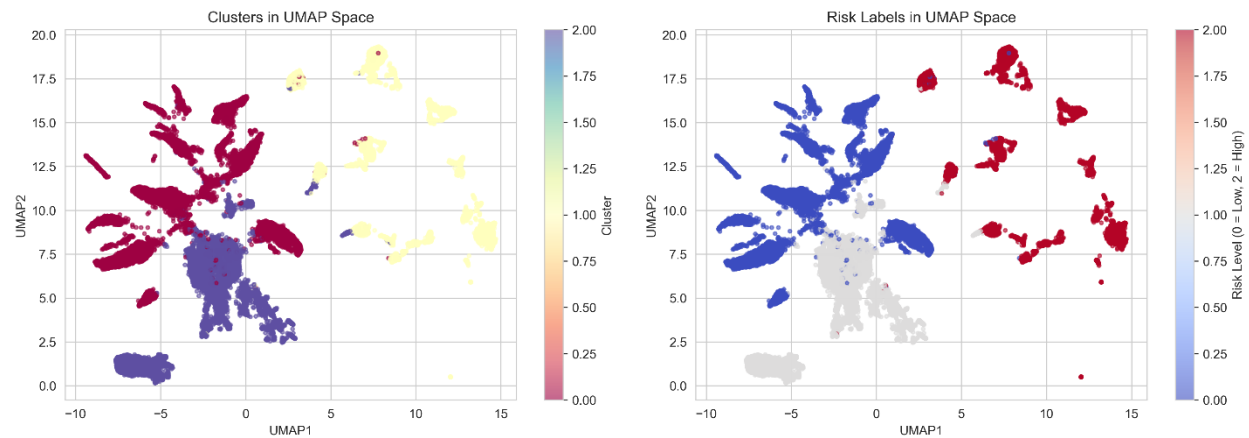


Figure 6

Figure 6 presents the GMM clustering result in UMAP space. Because GMM models probabilistic components, it can represent overlapping groups rather than forcing only hard, equally shaped partitions. The project contains 35 visualizations, available in the figures folder.

4. Plot Analysis

This section presents all 35 project visualizations in the same order in which they are generated by the Python workflow. Each figure is followed by a concise interpretation focused on its analytical meaning and its role in the credit-risk study.

4.1 Bill Amount Distributions

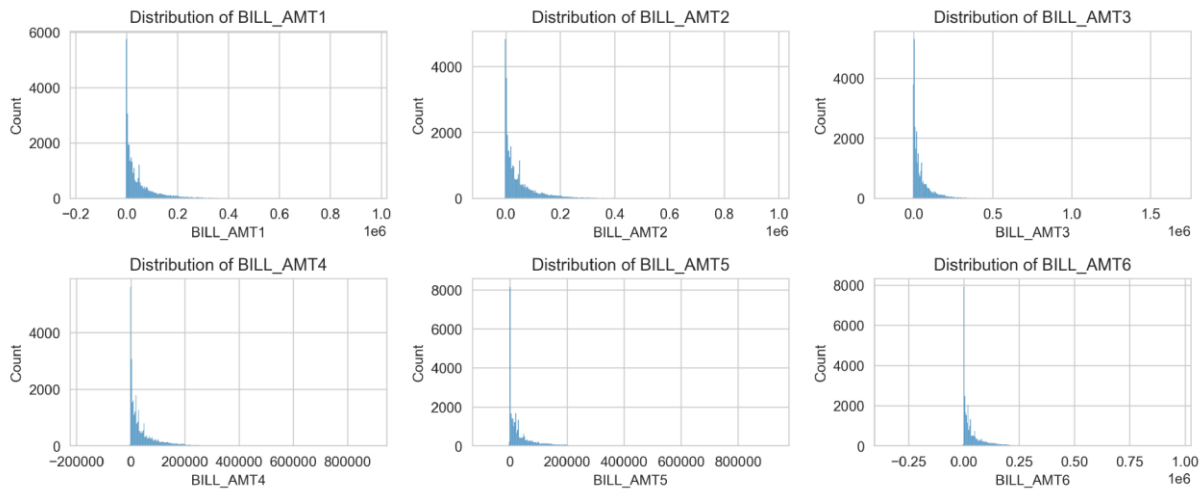


Figure 7. Bill Amount Distributions

All six bill-amount variables are strongly right-skewed. Most customers are concentrated at relatively low or moderate bill values, while a small number of very large balances create long upper tails. This pattern shows that the financial variables are not normally distributed and supports the use of outlier-resistant preprocessing.

4.2 Payment Amount Distributions

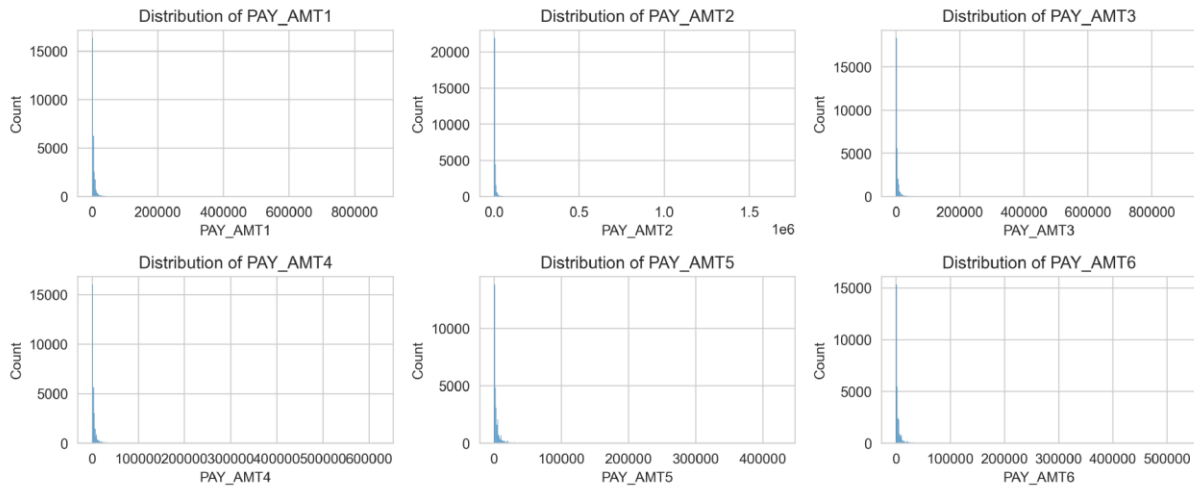


Figure 8. Payment Amount Distributions

Payment amounts are even more concentrated near zero, with a limited number of very large payments extending the right tail. The same general shape appears across all six months. This indicates that ordinary mean-based scaling could be strongly affected by a small group of customers with unusually large payments.

4.3 Bill Amount Boxplots

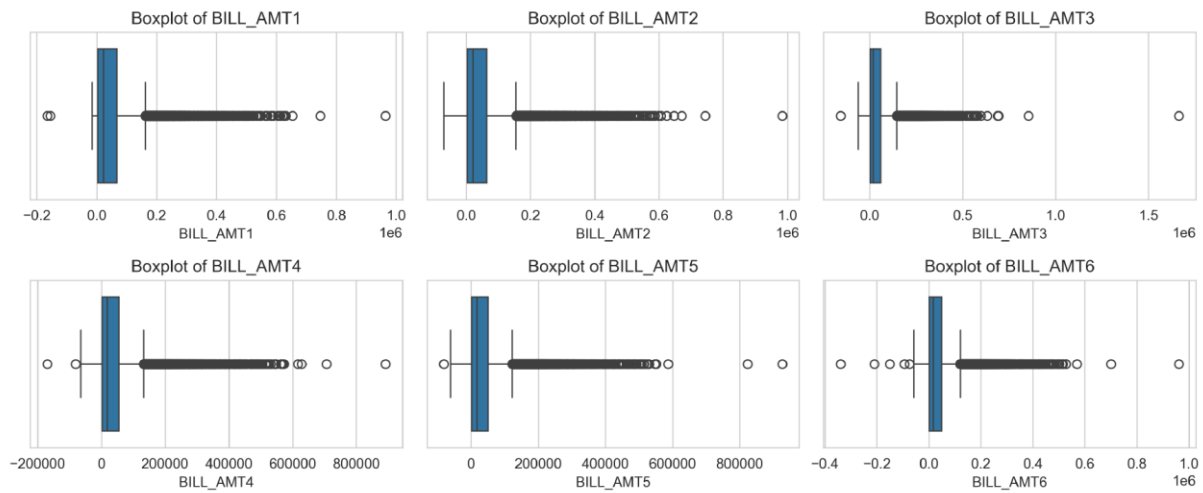


Figure 9. Bill Amount Boxplots

The bill-amount boxplots confirm the presence of many high-value observations outside the interquartile range. These points are not automatically treated as errors because large bills can be valid, but their influence must be controlled. Winsorization and RobustScaler are therefore suitable choices for the continuous financial variables.

4.4 Payment Amount Boxplots

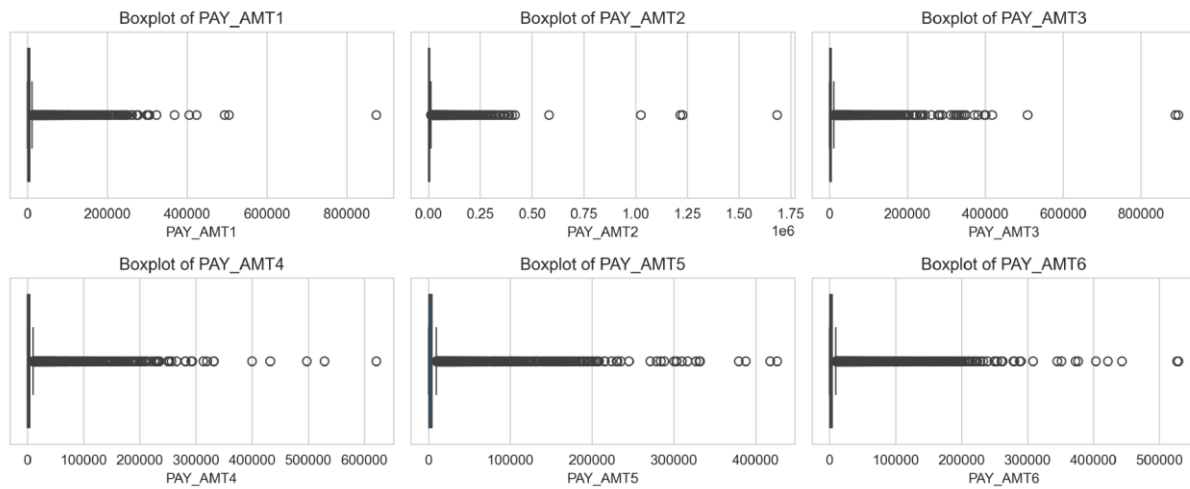


Figure 10. Payment Amount Boxplots

The payment boxplots show narrow central ranges and a large number of upper-tail values. Many customers make small payments, while a much smaller group makes exceptionally large payments. This imbalance explains the high skewness and kurtosis reported for the payment variables.

4.5 Target Class Distribution



Figure 11. Target Class Distribution

The non-default class is much larger than the default class: 23,364 customers did not default and 6,636 did. The default class represents about 22.12% of the sample. Because of this imbalance, accuracy alone is not enough and the evaluation must also consider recall, precision, F1, ROC-AUC, and Average Precision.

4.6 Customers by Age Group

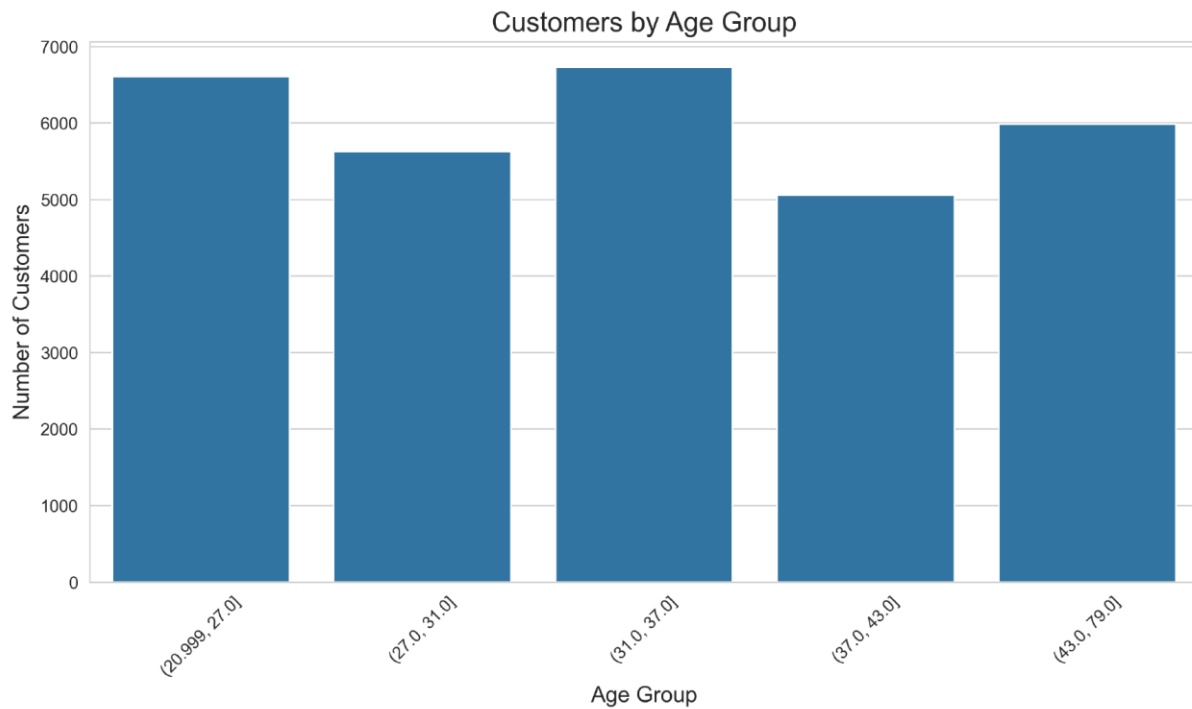


Figure 12. Customers by Age Group

The five age groups contain broadly similar numbers of customers because they were created with quantile-based grouping. Small differences remain because customers with the same age cannot always be divided across separate intervals. The figure is useful for comparing groups, but it should not be interpreted as evidence that age itself is uniformly distributed.

4.7 Late Payment Rate by Marital Status and Sex

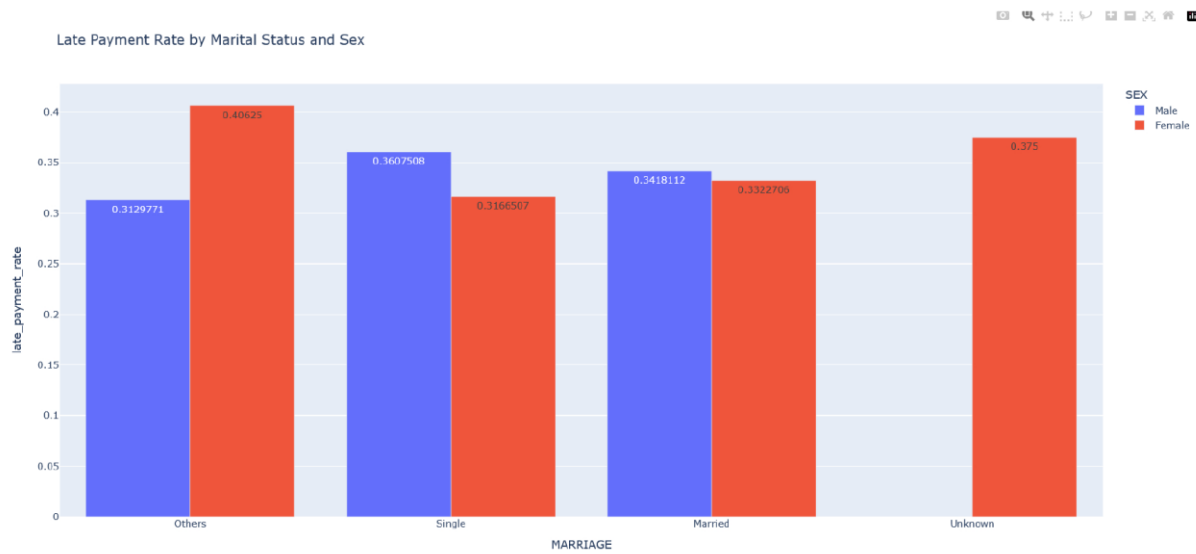


Figure 13. Late Payment Rate by Marital Status and Sex

Late-payment rates vary across the marital-status and sex combinations. For the two largest marital groups, men generally show higher late-payment rates than women, although the differences are not equally large in every category. The small Unknown and Other groups should be interpreted cautiously because their sample sizes are limited.

4.8 Repayment Status by Default Class

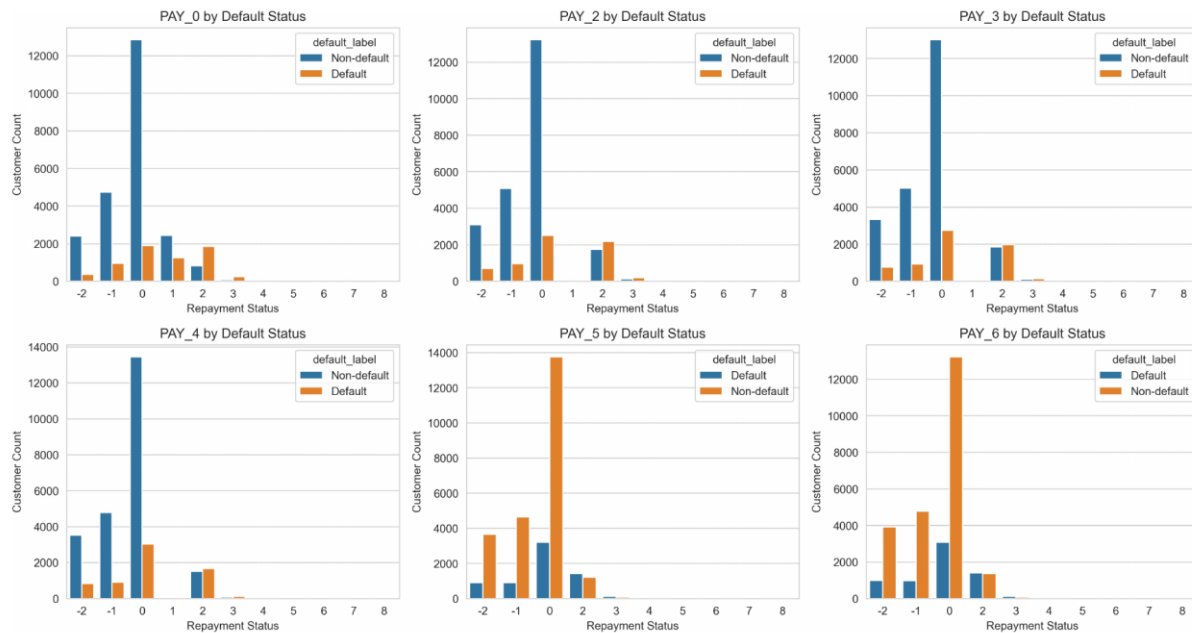


Figure 14. Repayment Status by Default Class

Recent repayment-status variables provide clear separation between the two target classes. Defaulting customers appear more frequently in positive delay categories, especially in the most recent months. The difference becomes weaker for older months, which confirms that recent payment behavior is more useful for an early-warning system.

4.9 Credit Limit Summary by Default Status

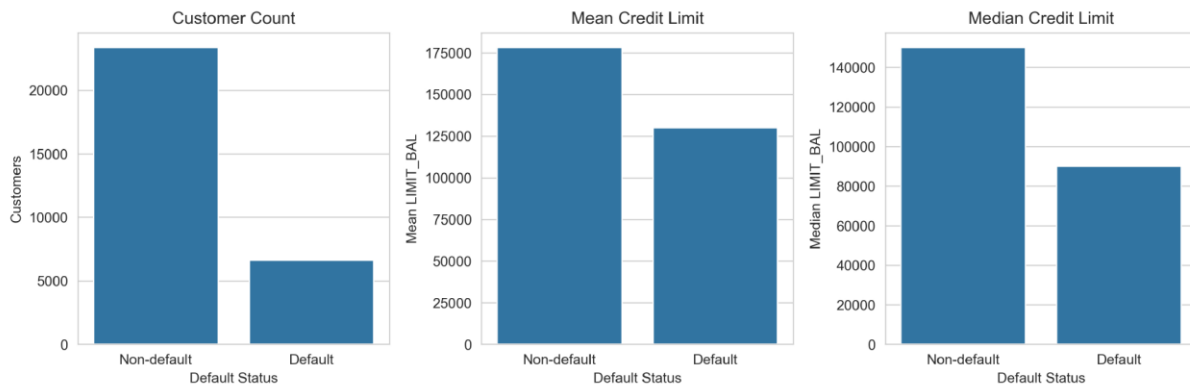


Figure 15. Credit Limit Summary by Default Status

Non-defaulting customers are more numerous and also have higher mean and median credit limits than defaulting customers. This suggests that customers with stronger credit profiles tend to receive larger limits. However, the difference in credit limit alone is not sufficient for classification and must be combined with repayment behavior and other variables.

4.10 Credit Limit Boxplot by Default Status

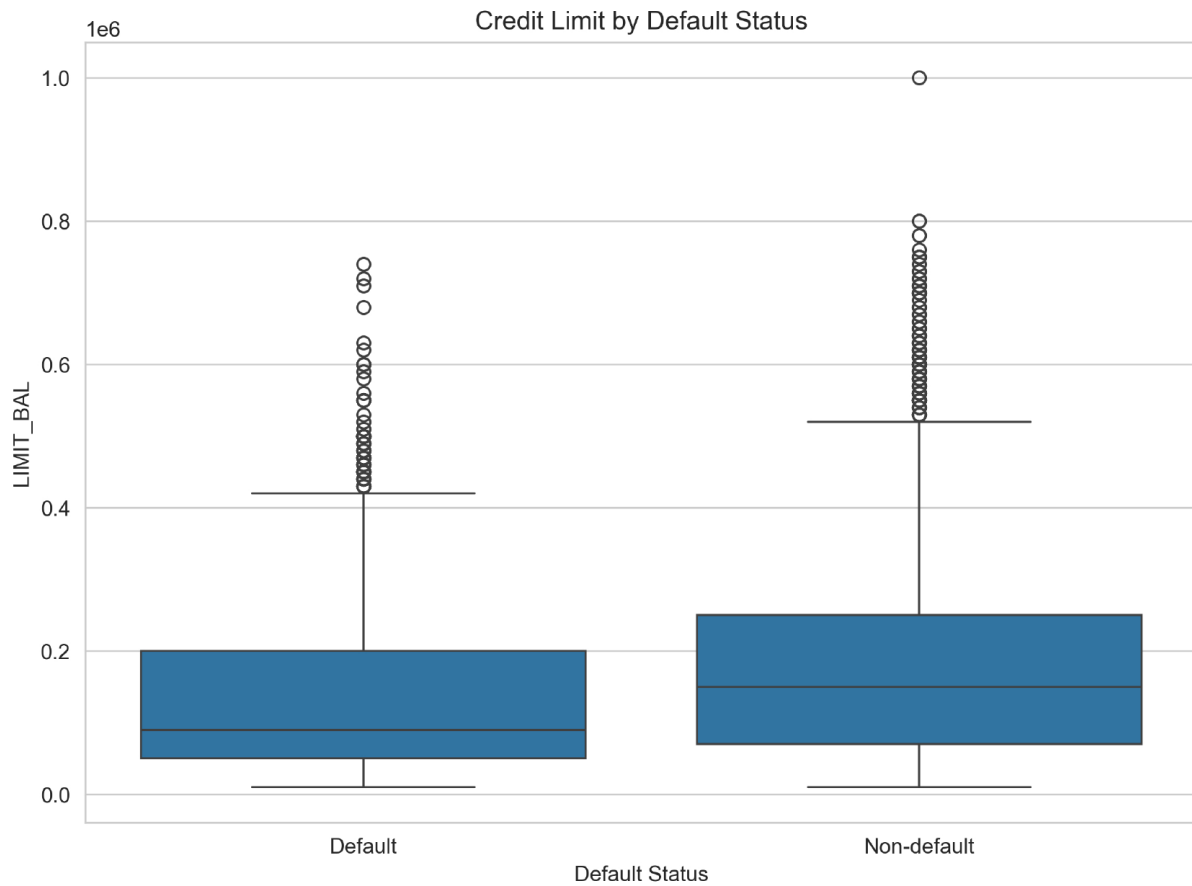


Figure 16. Credit Limit Boxplot by Default Status

The non-default group has a higher central credit-limit distribution, while both classes contain high-limit outliers. Considerable overlap remains between the groups, so credit limit cannot independently determine default. Its value is mainly as one component of a multivariable risk model.

4.11 Correlation Matrix of Numeric Features

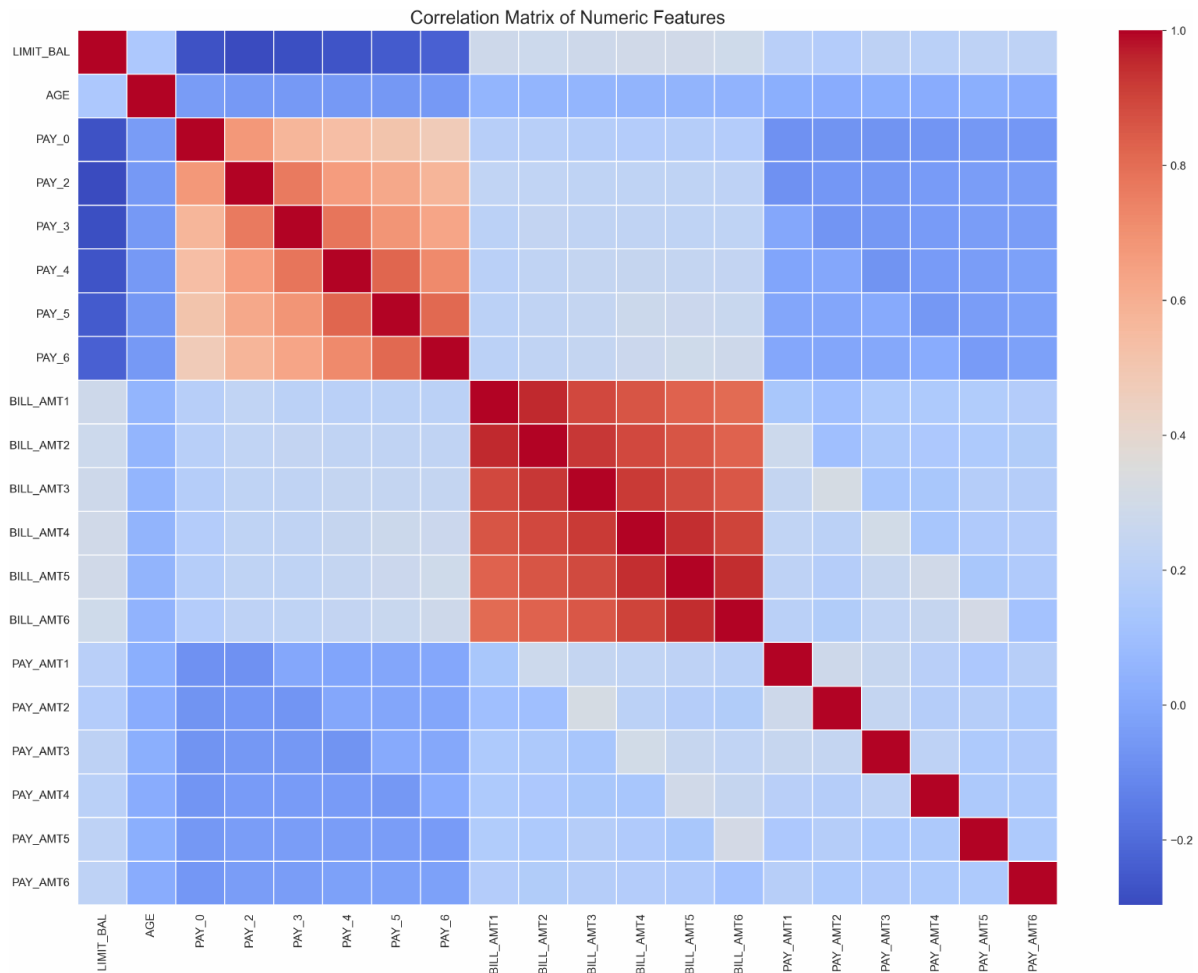


Figure 17. Correlation Matrix of Numeric Features

Monthly bill amounts are strongly correlated with one another because customer balances tend to persist over time. Repayment-status variables also show positive correlations across consecutive months. Payment amounts have weaker relationships with most variables, and the target does not have a single extremely strong linear correlation, supporting the use of several predictors together.

4.12 Repayment, Payment, and Bill Separation

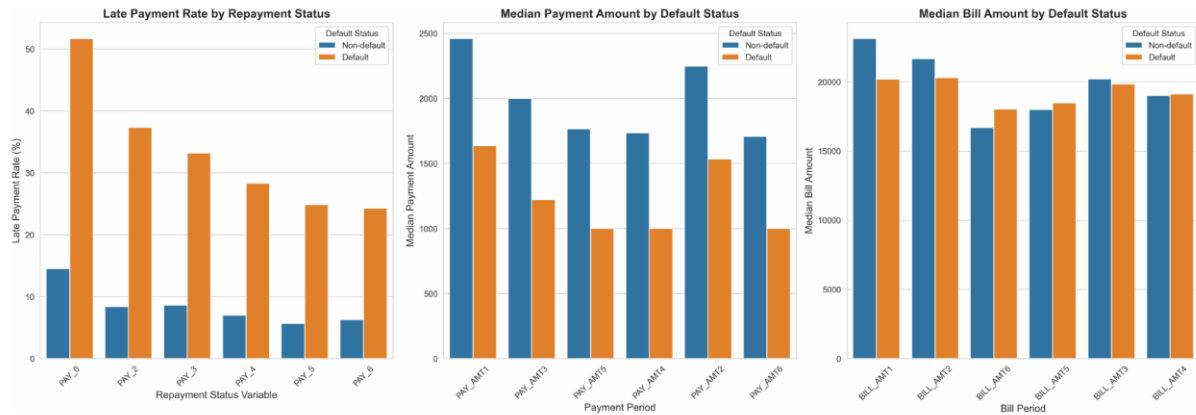


Figure 18. Repayment, Payment, and Bill Separation

The clearest difference between default and non-default customers comes from repayment status, particularly PAY_0. The current-month delay rate is 51.67% for defaulters and 14.51% for non-defaulters. Median payment amounts are consistently lower for the default group, whereas median bill amounts provide less stable separation.

4.13 Median Payment Trend by Default Status

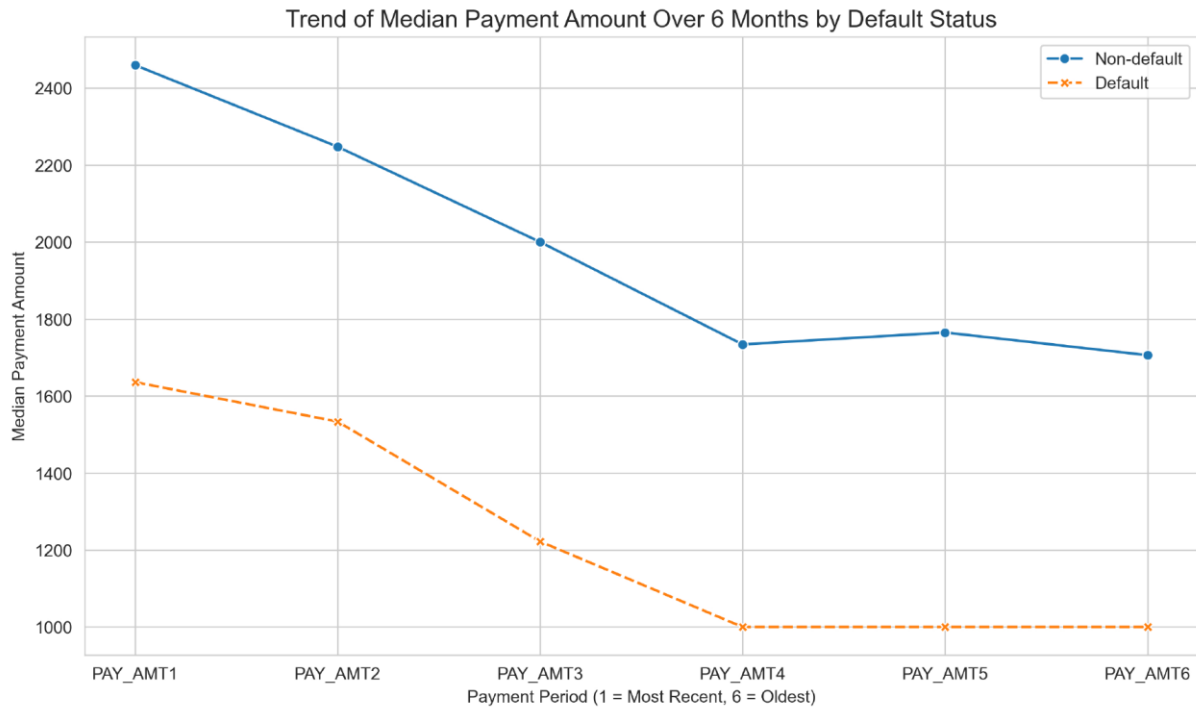


Figure 19. Median Payment Trend by Default Status

Across all six periods, non-defaulting customers have higher median payment amounts than defaulting customers. The gap is especially visible in the more recent periods and remains present throughout the six-month history. Consistently lower payments therefore act as an additional warning signal when combined with repayment delays.

4.14 Repayment-Ratio Threshold Sensitivity

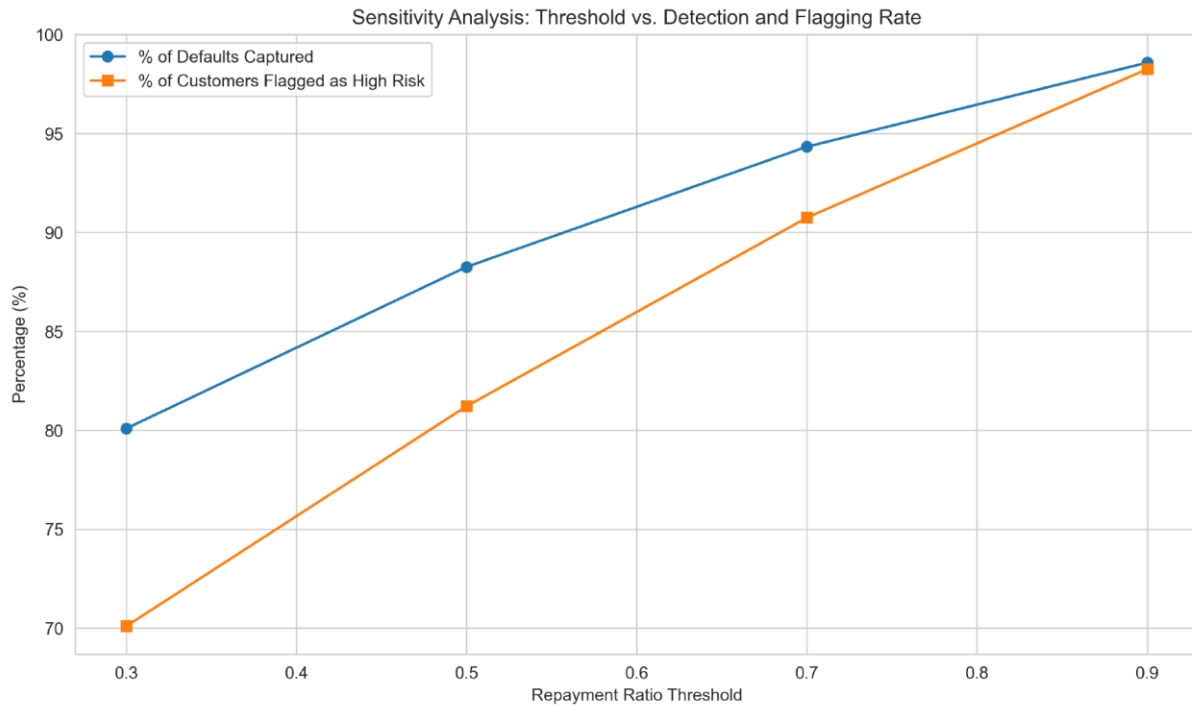


Figure 20. Repayment-Ratio Threshold Sensitivity

Raising the repayment-ratio threshold captures a larger percentage of defaults, but it also labels a much larger share of customers as high risk. For example, the 0.30 threshold captures about 80.09% of defaults while flagging 70.10% of customers; at 0.90, both values rise to about 98%. This shows why a threshold must balance detection against operational workload.

4.15 PAY_0 and Marital Status Interaction

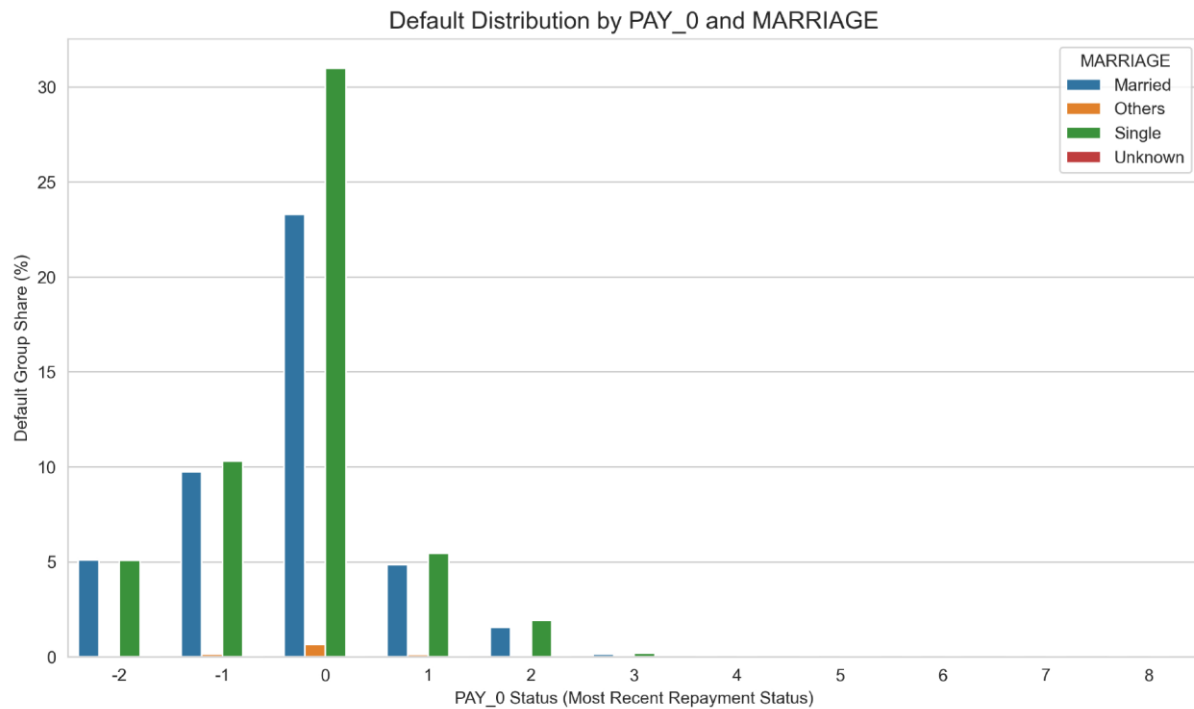


Figure 21. PAY_0 and Marital Status Interaction

The strongest default shares occur around PAY_0 values of 0, 1, and 2, mainly because these categories contain most observations. The sharp increase for delayed categories is visible across marital groups, especially at PAY_0 = 2. Repayment status is therefore more informative than marital status, while their interaction can provide additional context.

4.16 PCA Biplot of Payment Amounts

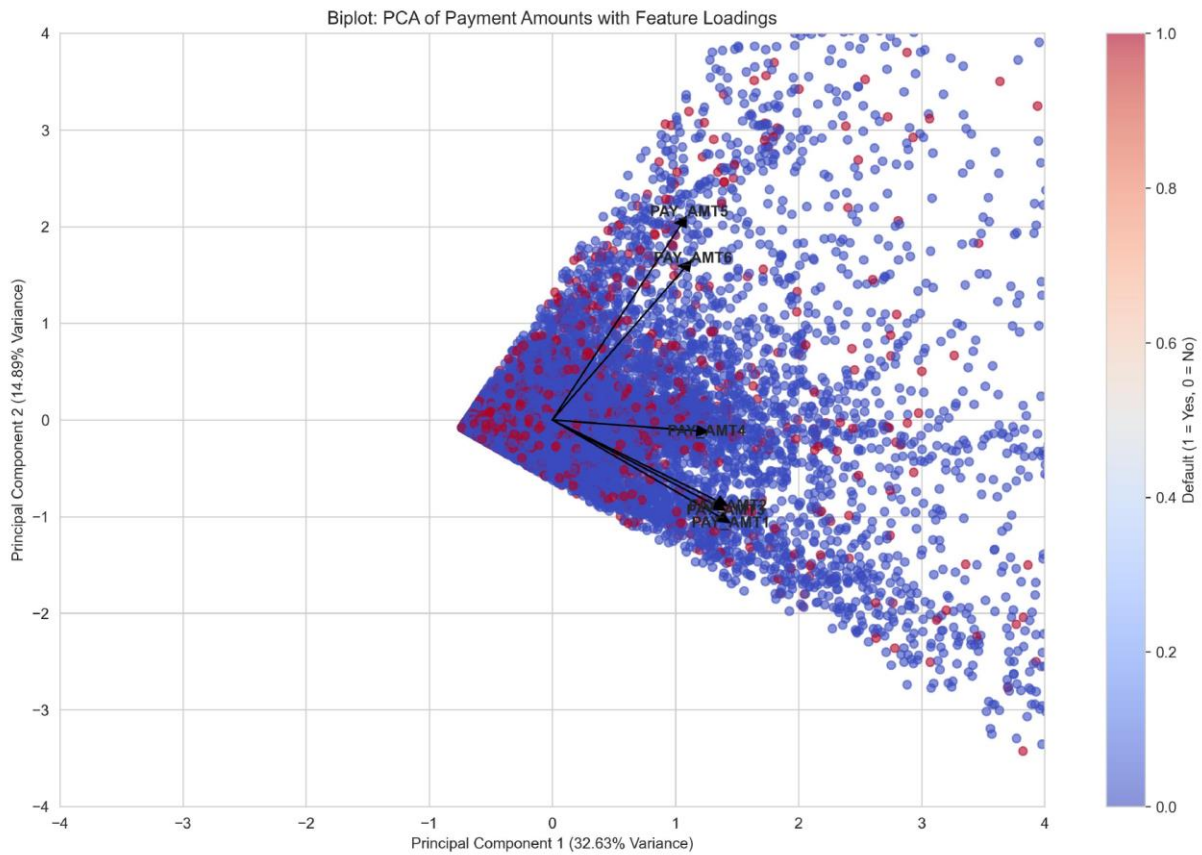


Figure 22. PCA Biplot of Payment Amounts

The first principal component loads positively on all six payment variables and represents a general level of payment activity. The second component is more strongly influenced by PAY_AMT5 and PAY_AMT6 in the positive direction and by recent payments in the negative direction. Default and non-default observations still overlap, so the PCA projection is descriptive rather than a direct classifier.

4.17 Distributions Before Preprocessing

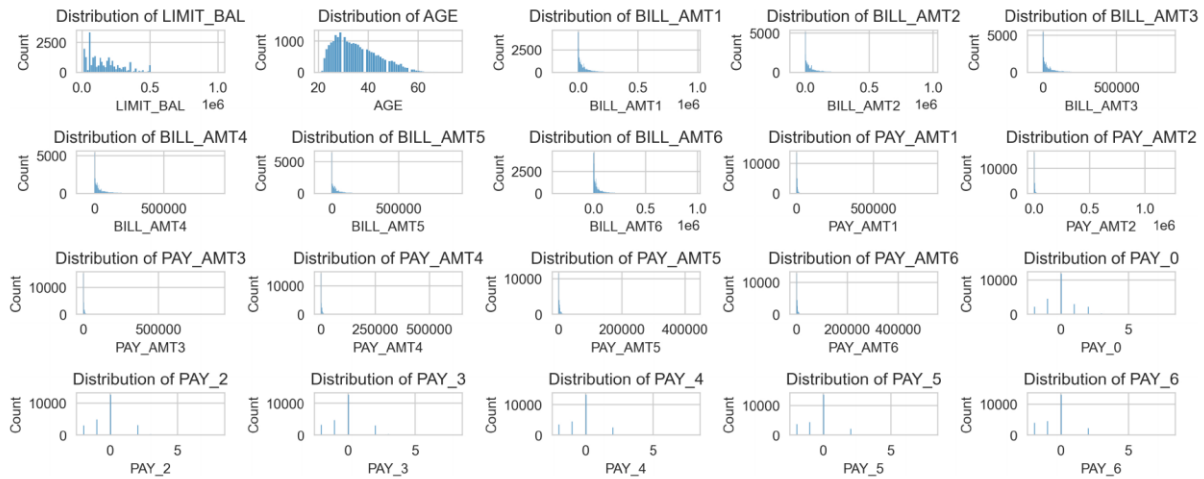


Figure 23. Distributions Before Preprocessing

Before preprocessing, continuous financial variables are heavily right-skewed and repayment-status variables are discrete. Several features have long tails and very different numerical scales. These conditions can reduce the stability of a linear model if raw values are used directly.

4.18 Boxplots Before Preprocessing

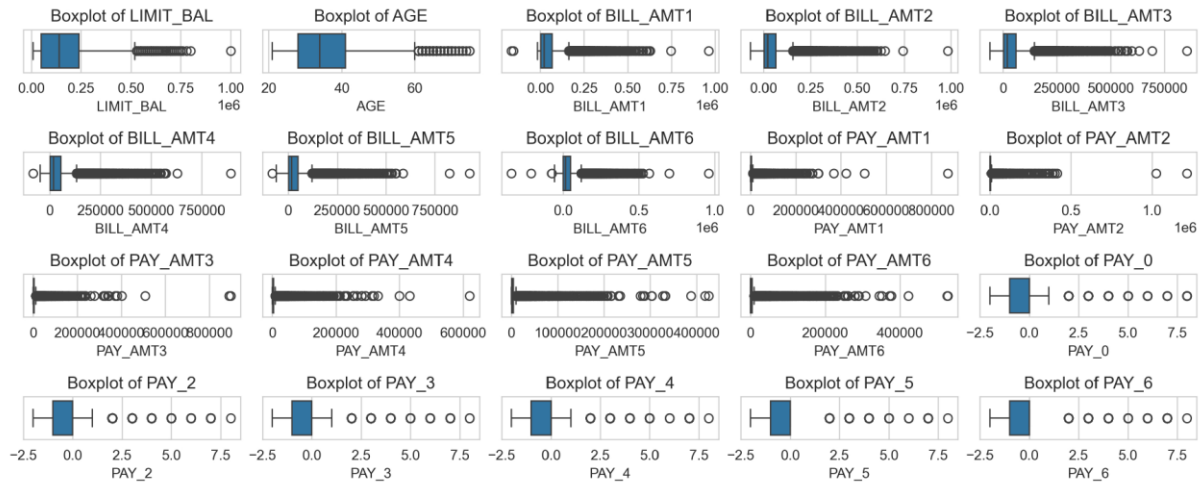


Figure 24. Boxplots Before Preprocessing

The raw continuous variables contain many extreme observations, particularly bill and payment amounts. The repayment-status variables also appear as separated discrete levels rather than continuous measurements. This figure provides the visual reason for applying the numeric transformation only to continuous columns while passing ordinal PAY variables through separately.

4.19 Distributions After Preprocessing

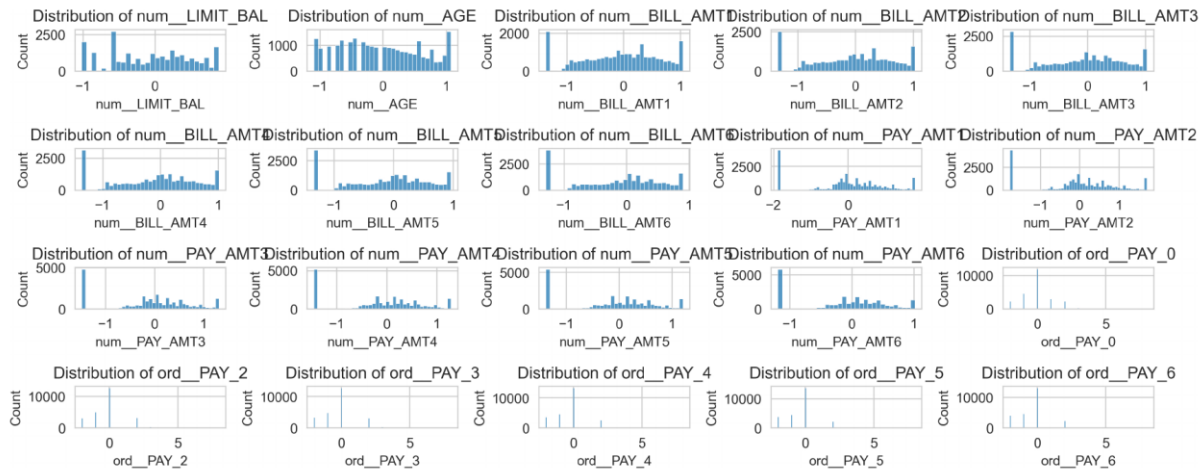


Figure 25. Distributions After Preprocessing

After winsorization, Yeo-Johnson transformation, and RobustScaler, the continuous variables are more centered and their extreme tails are reduced. The transformation improves scale comparability and numerical stability, but it does not guarantee perfect normality. Ordinal repayment-status variables remain discrete because they are intentionally passed through unchanged.

4.20 Boxplots After Preprocessing

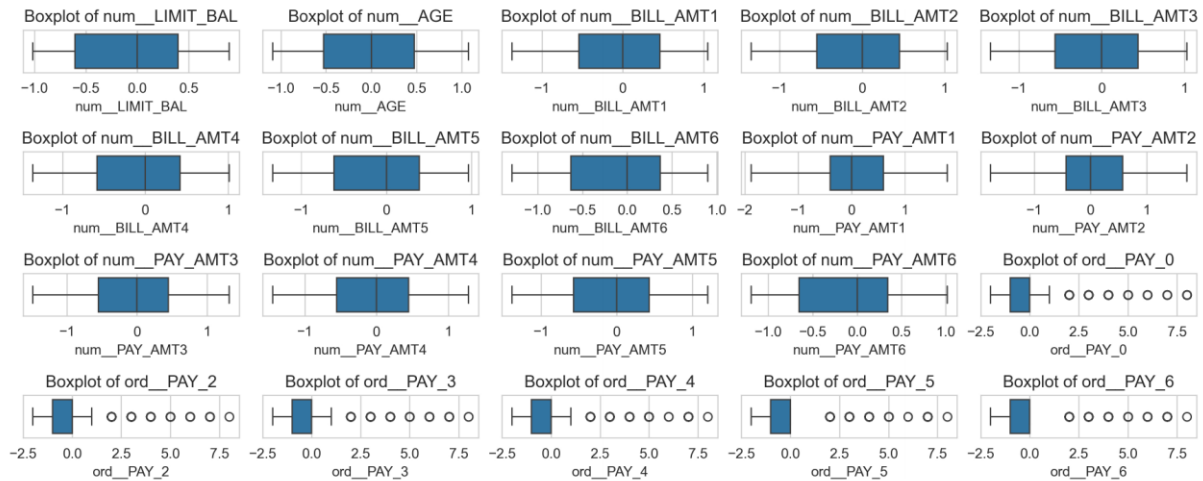


Figure 26. Boxplots After Preprocessing

The transformed continuous features have more controlled ranges and reduced outlier influence compared with the raw data. Some apparent extreme points remain because the distributions are still not perfectly symmetric and the PAY variables were not winsorized. The goal of preprocessing is robust modeling, not the complete removal of every statistical outlier.

4.21 Logistic Regression Confusion Matrix

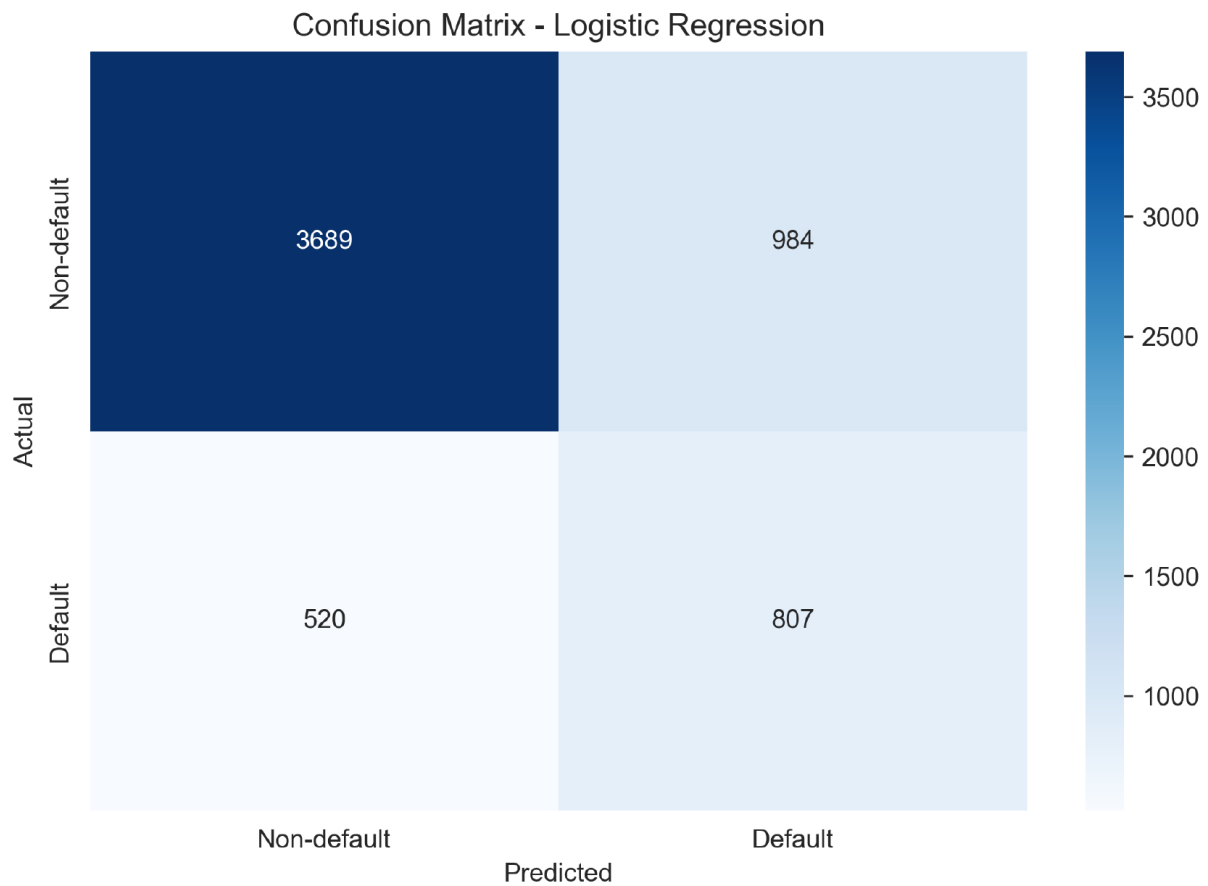


Figure 27. Logistic Regression Confusion Matrix

At the standard 0.50 threshold, the optimized Logistic Regression correctly identifies 807 defaulters and 3,689 non-defaulters. It misses 520 defaults and incorrectly flags 984 non-default customers. Compared with the simple baseline, SMOTENC substantially reduces missed defaults, although it creates more false alerts.

4.22 Logistic Regression ROC Curve

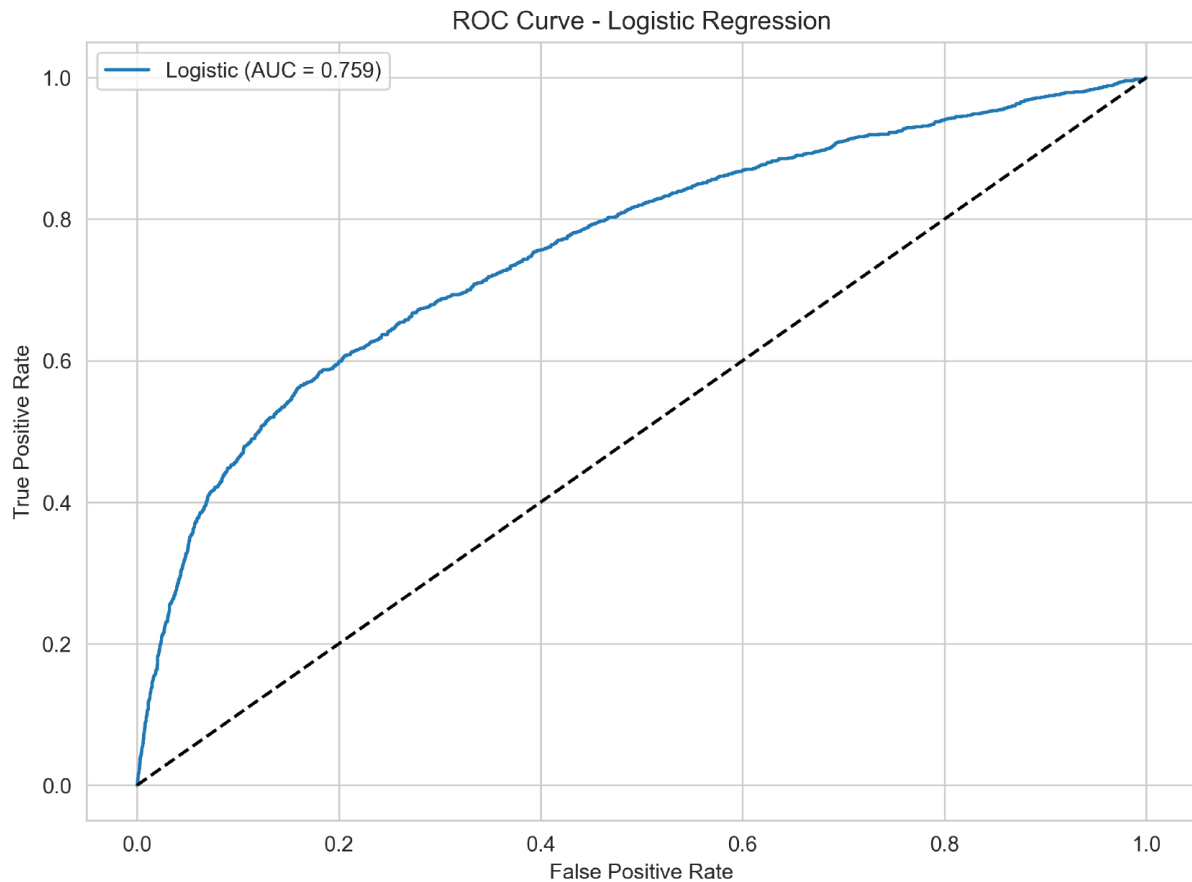


Figure 28. Logistic Regression ROC Curve

The Logistic Regression ROC-AUC is 0.759, showing a useful but not perfect ability to rank defaulting customers above non-defaulting customers across possible thresholds. The curve remains clearly above the diagonal random-classifier line. The result should be considered together with Precision-Recall performance because the target is imbalanced.

4.23 Logistic Regression Precision-Recall Curve

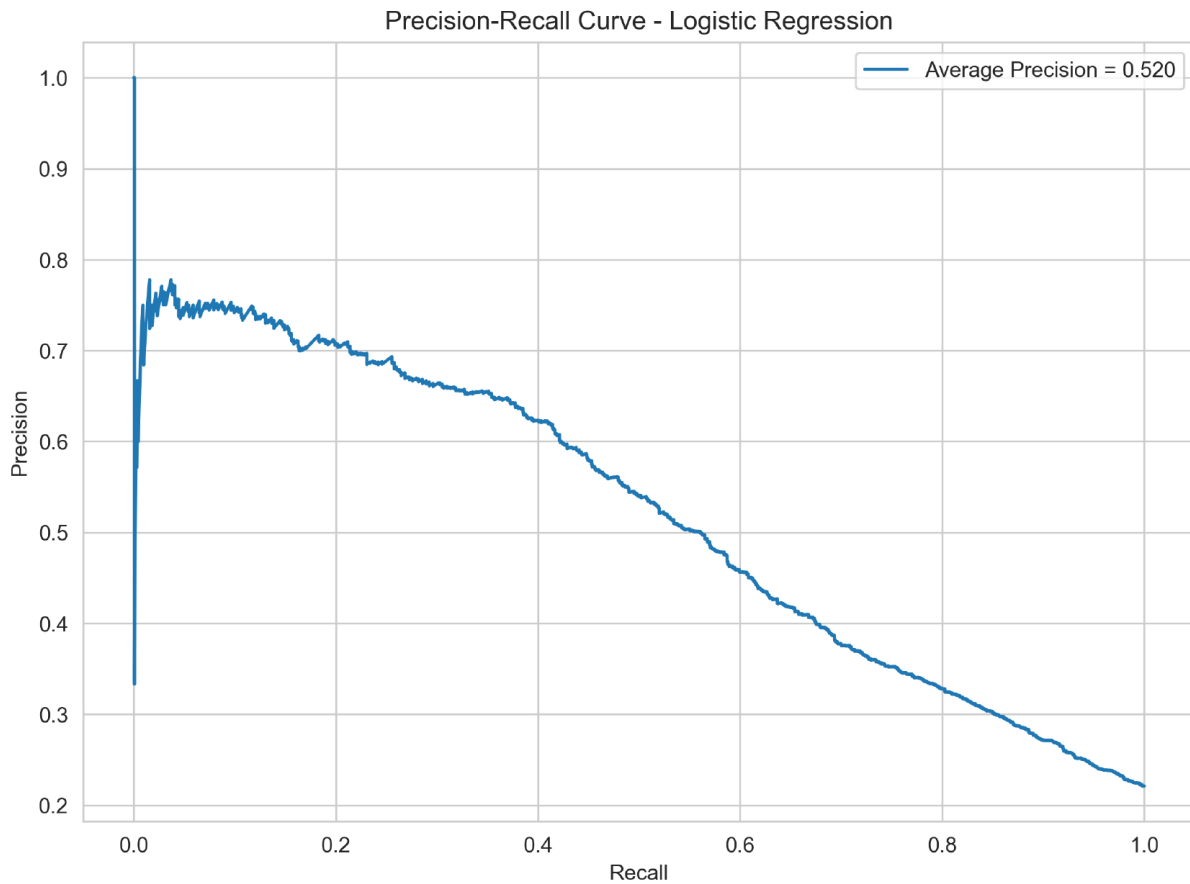


Figure 29. Logistic Regression Precision-Recall Curve

The Average Precision is 0.520, which is meaningfully higher than the default prevalence of about 0.221. Precision gradually declines as recall approaches one, illustrating the cost of trying to detect nearly every default. The operating point should therefore be selected according to the cost of missed defaults and false alerts.

4.24 Top Logistic Regression Coefficients

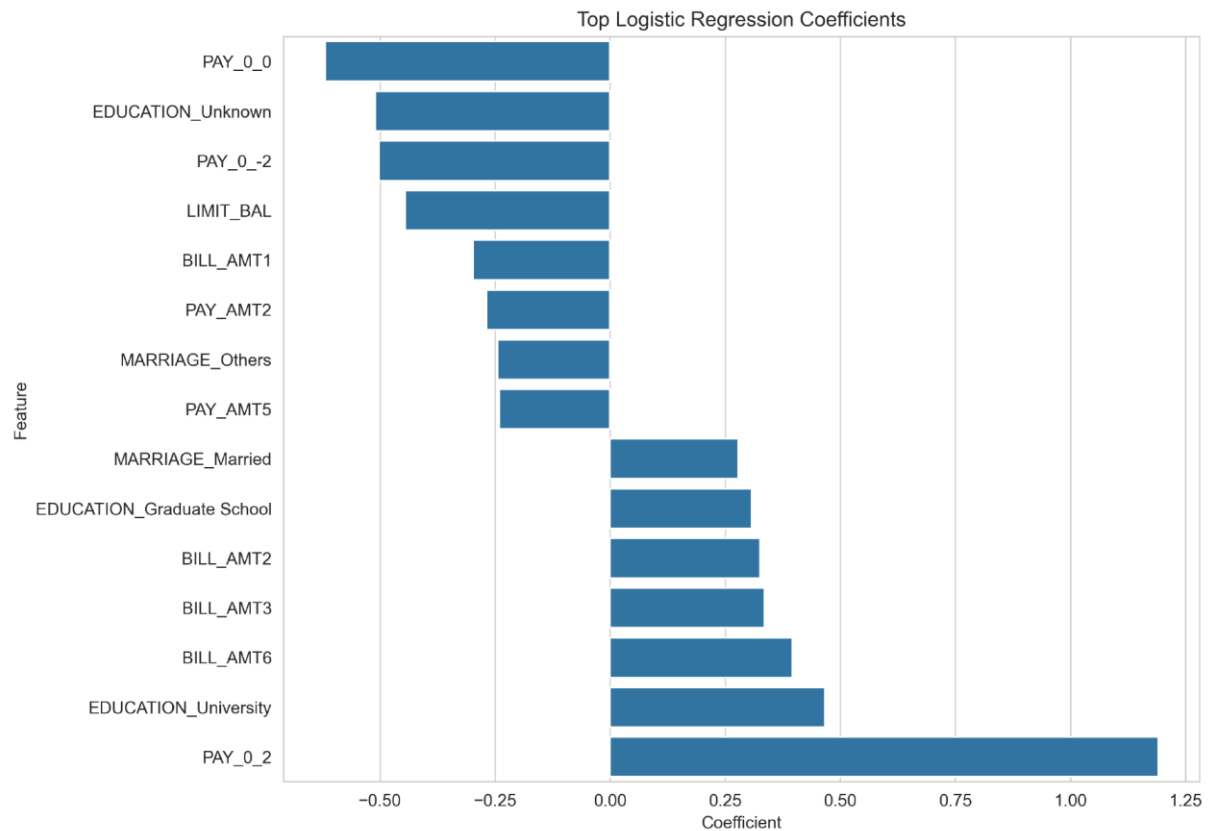


Figure 30. Top Logistic Regression Coefficients

PAY_0 = 2 has the strongest positive coefficient and is the clearest indicator of increased predicted default risk. Higher LIMIT_BAL and PAY_AMT2 are associated with lower predicted risk, while several bill amounts have positive coefficients. Coefficients describe conditional model relationships after preprocessing; they should not be interpreted as proof of causation.

4.25 Logistic Regression SHAP Summary

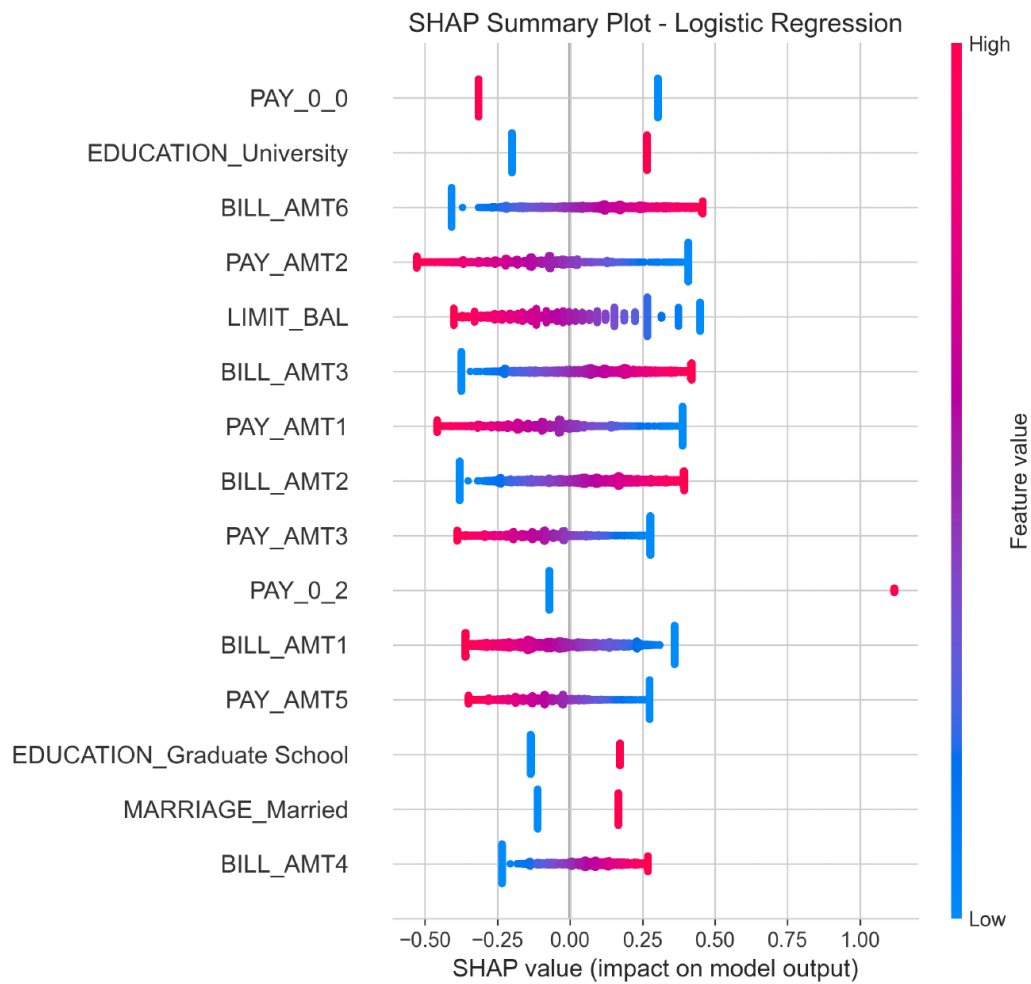


Figure 31. Logistic Regression SHAP Summary

The SHAP summary shows both the importance and direction of feature effects across customers. PAY_0, education categories, bill amounts, payment amounts, and credit limit contribute most strongly. Higher payment amounts and higher credit limits generally reduce predicted risk, while recent delays and some higher bill values increase it.

4.26 Logistic Regression SHAP Feature Importance

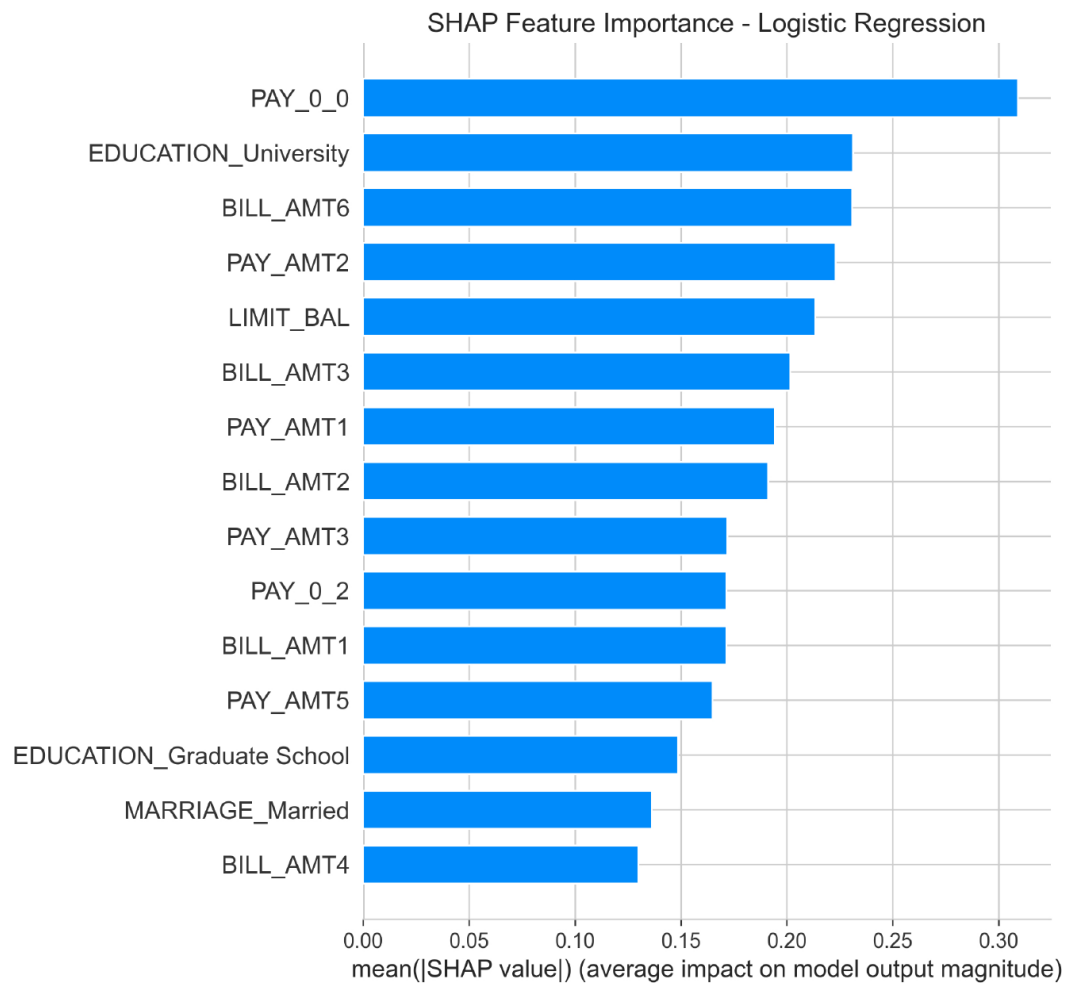


Figure 32. Logistic Regression SHAP Feature Importance

The mean absolute SHAP ranking places PAY_0, university education, BILL_AMT6, PAY_AMT2, and LIMIT_BAL among the leading predictors. This ranking measures average prediction impact without showing direction. Its agreement with the coefficient and RFE results strengthens the conclusion that recent repayment behavior and financial capacity are central to default prediction.

4.27 PCA, UMAP, and t-SNE Comparison

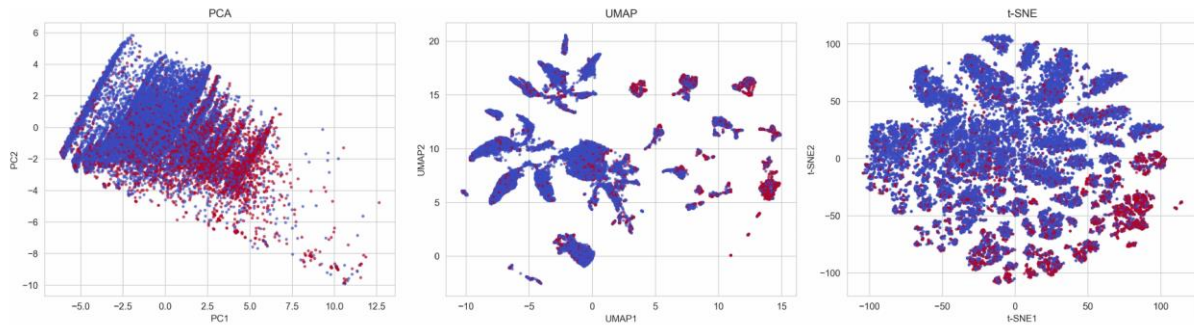


Figure 33. PCA, UMAP, and t-SNE Comparison

PCA shows substantial overlap because a linear two-dimensional projection cannot fully represent the structure of the transformed data. UMAP produces clearer regions, while t-SNE reveals several local groups. Default and non-default points remain mixed in all three views, indicating that customer risk is not separated by one simple two-dimensional boundary.

4.28 UMAP Clusters and Risk Labels

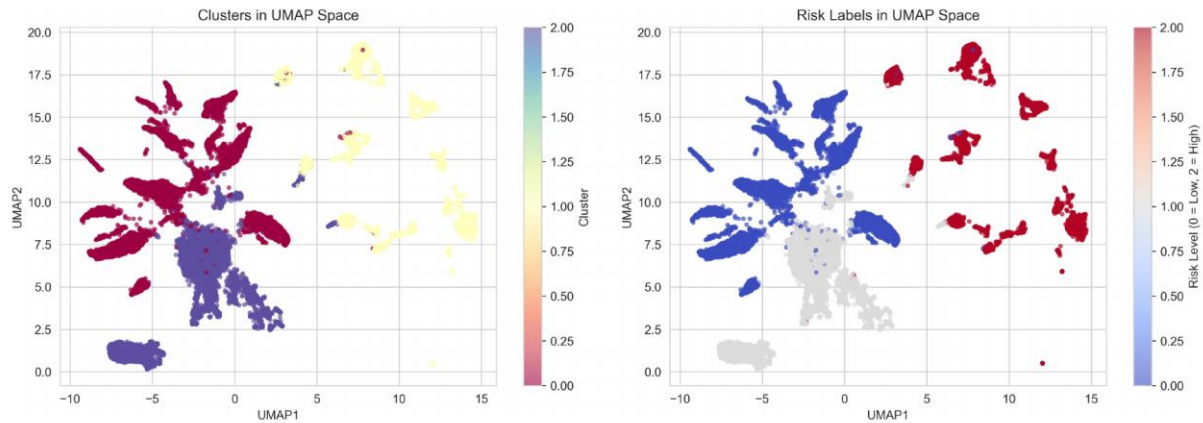


Figure 34. UMAP Clusters and Risk Labels

The left panel displays the three GMM clusters in UMAP space, and the right panel translates those numeric clusters into Low, Medium, and High Risk labels. The clusters form visible regions with some boundary overlap, which is consistent with the probabilistic nature of GMM. Risk names were assigned using training-set default rates, not used to create the clusters.

4.29 PAY_0 and Payment Amount by Cluster

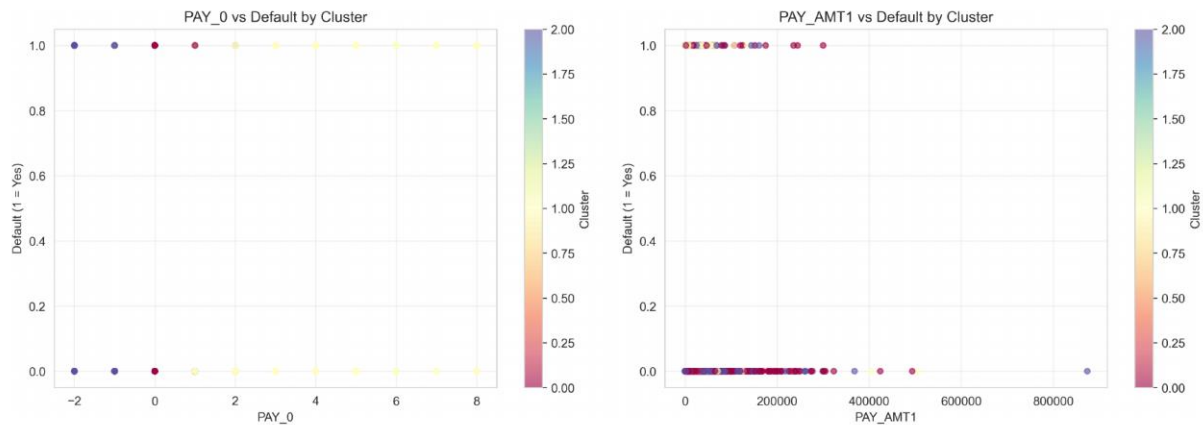


Figure 35. PAY_0 and Payment Amount by Cluster

The discrete PAY_0 plot shows that cluster membership is strongly connected to repayment delay patterns. The payment-amount panel also indicates that many High Risk observations are concentrated among customers with lower payments, although overlap remains. The horizontal bands occur because the default target is binary.

4.30 Random Forest Cluster Confusion Matrix

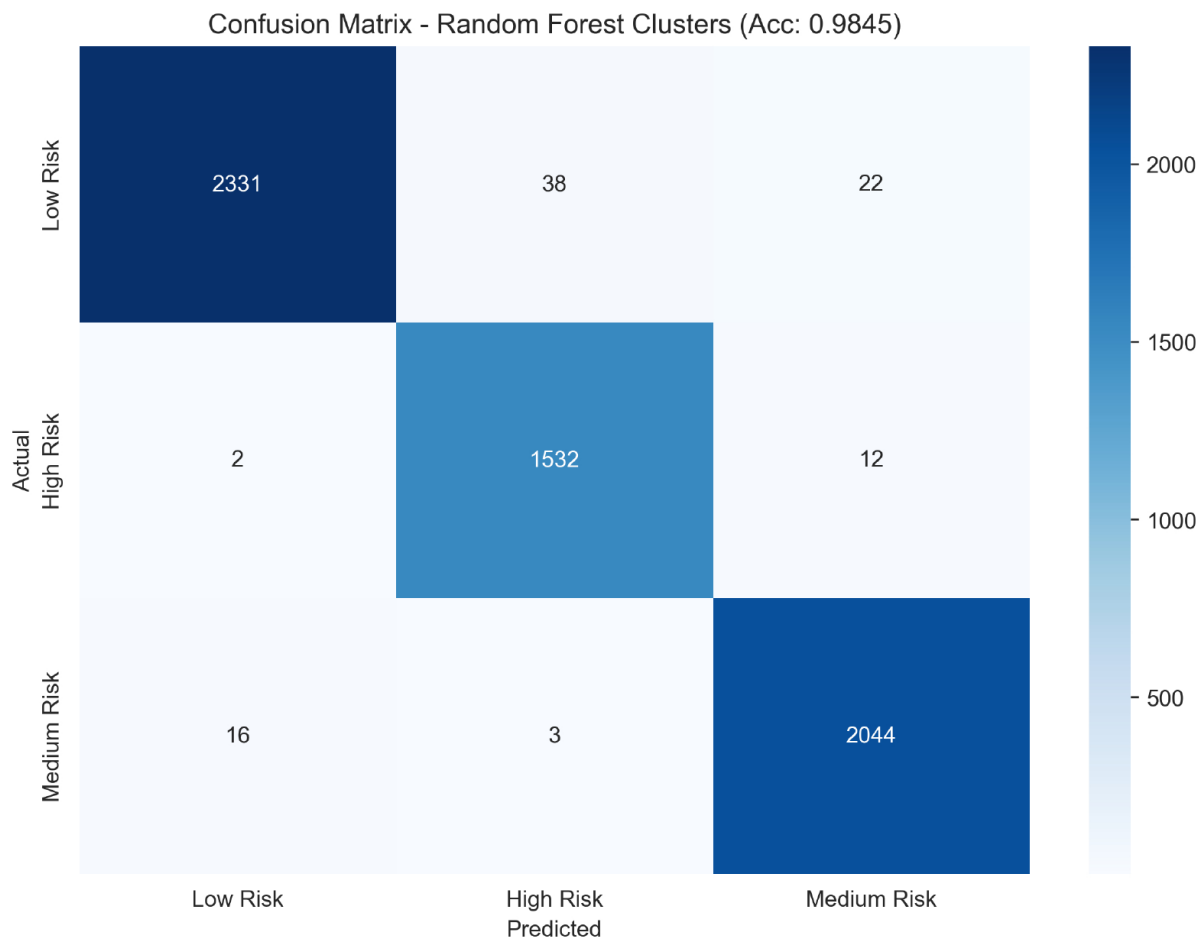


Figure 36. Random Forest Cluster Confusion Matrix

The Random Forest reproduces the GMM cluster labels with 98.45% test accuracy. Most errors occur near boundaries between the Low, Medium, and High Risk clusters, while the diagonal counts dominate the matrix. This is cluster-label prediction accuracy and must not be reported as default-prediction accuracy.

4.31 Random Forest ROC Curve for the High Risk Cluster

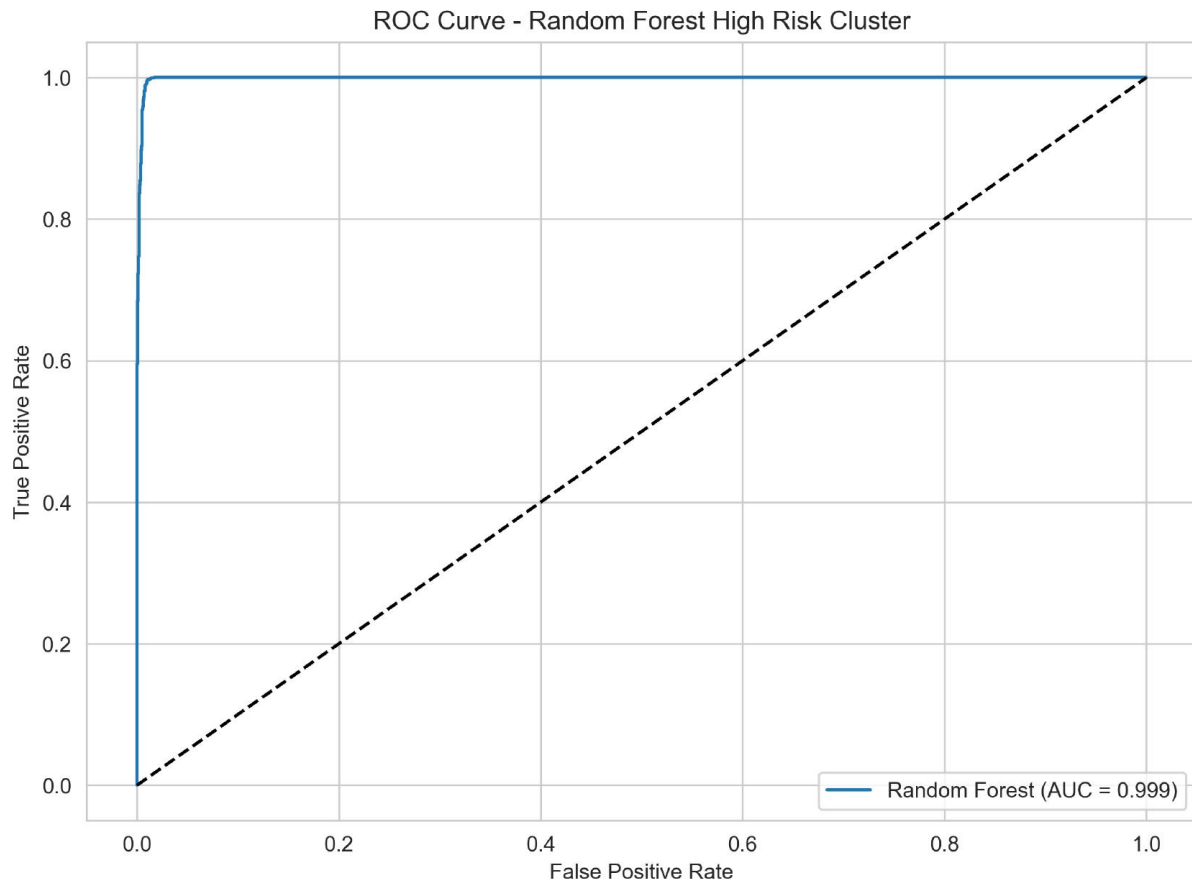


Figure 37. Random Forest ROC Curve for the High Risk Cluster

The one-versus-rest ROC curve for High Risk cluster membership has an AUC of approximately 0.999. This indicates that the Random Forest can almost perfectly rank High Risk cluster members relative to the other two clusters. The result is expected to be strong because the model is learning labels derived from the same predictor structure.

4.32 Random Forest Precision-Recall Curve for the High Risk Cluster

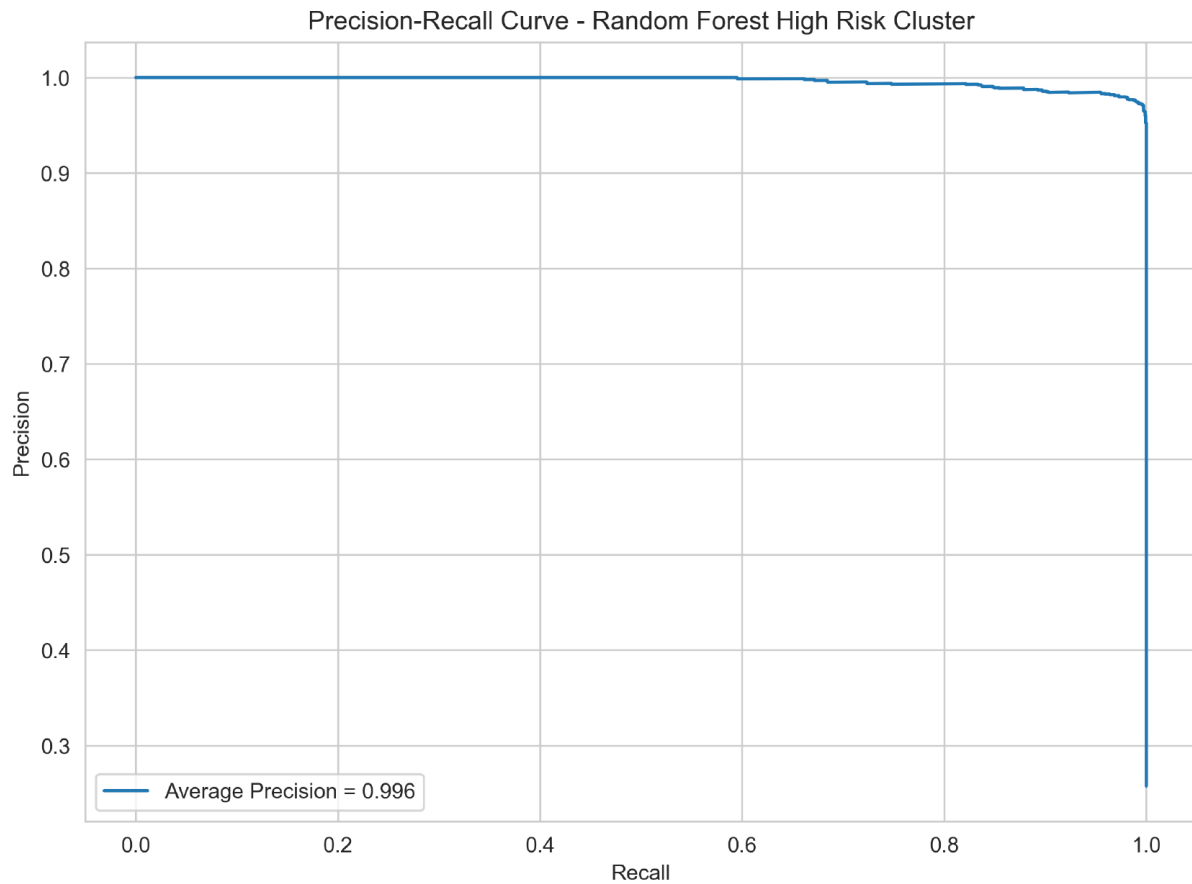


Figure 38. Random Forest Precision-Recall Curve for the High Risk Cluster

Average Precision is approximately 0.996 for identifying the High Risk cluster. Precision remains close to one across most recall values and falls mainly at the extreme end. Again, this evaluates agreement with the GMM-derived High Risk label rather than the occurrence of future default.

4.33 Random Forest SHAP Summary for the High Risk Cluster

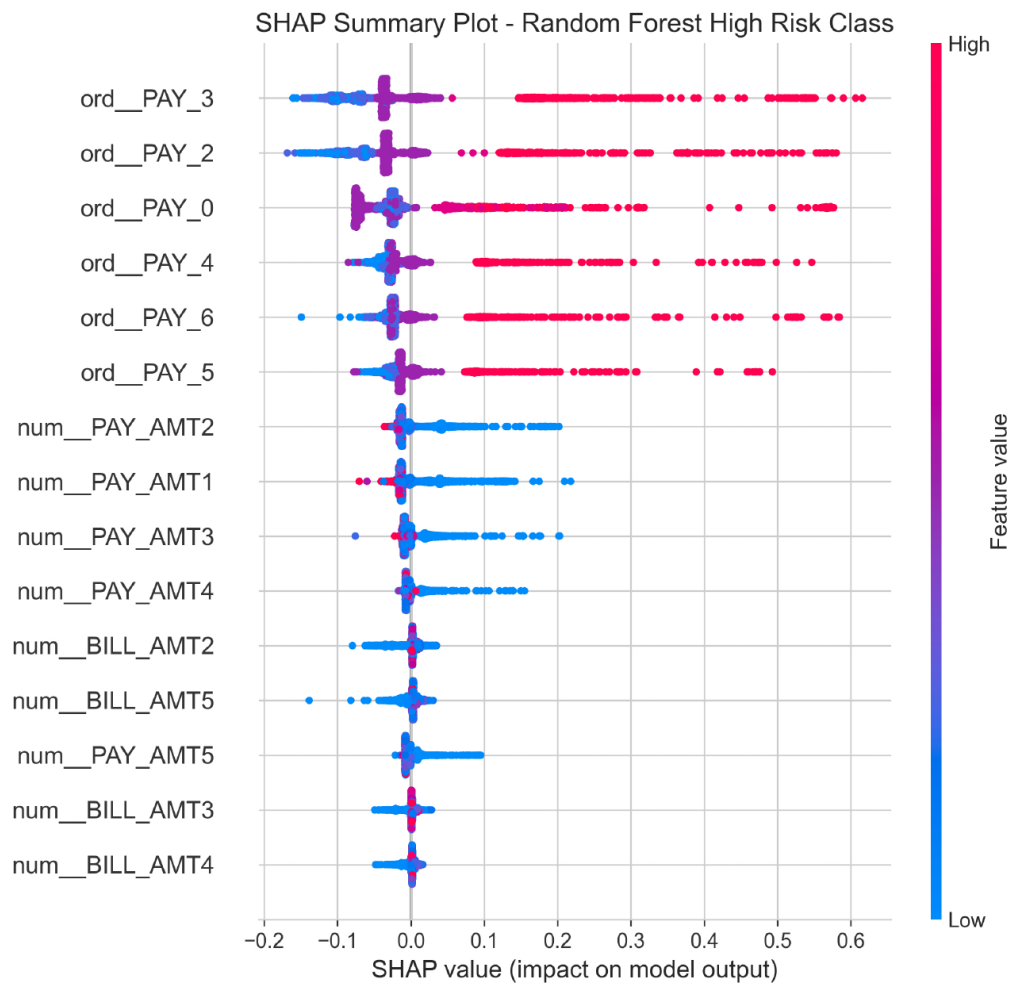


Figure 39. Random Forest SHAP Summary for the High Risk Cluster

High values of PAY_3, PAY_2, PAY_0, PAY_4, PAY_6, and PAY_5 generally increase the probability of High Risk cluster membership. Payment amounts have smaller but still visible effects. The plot confirms that repeated repayment delays across several months define the cluster more strongly than demographic variables.

4.34 Random Forest SHAP Feature Importance for the High Risk Cluster

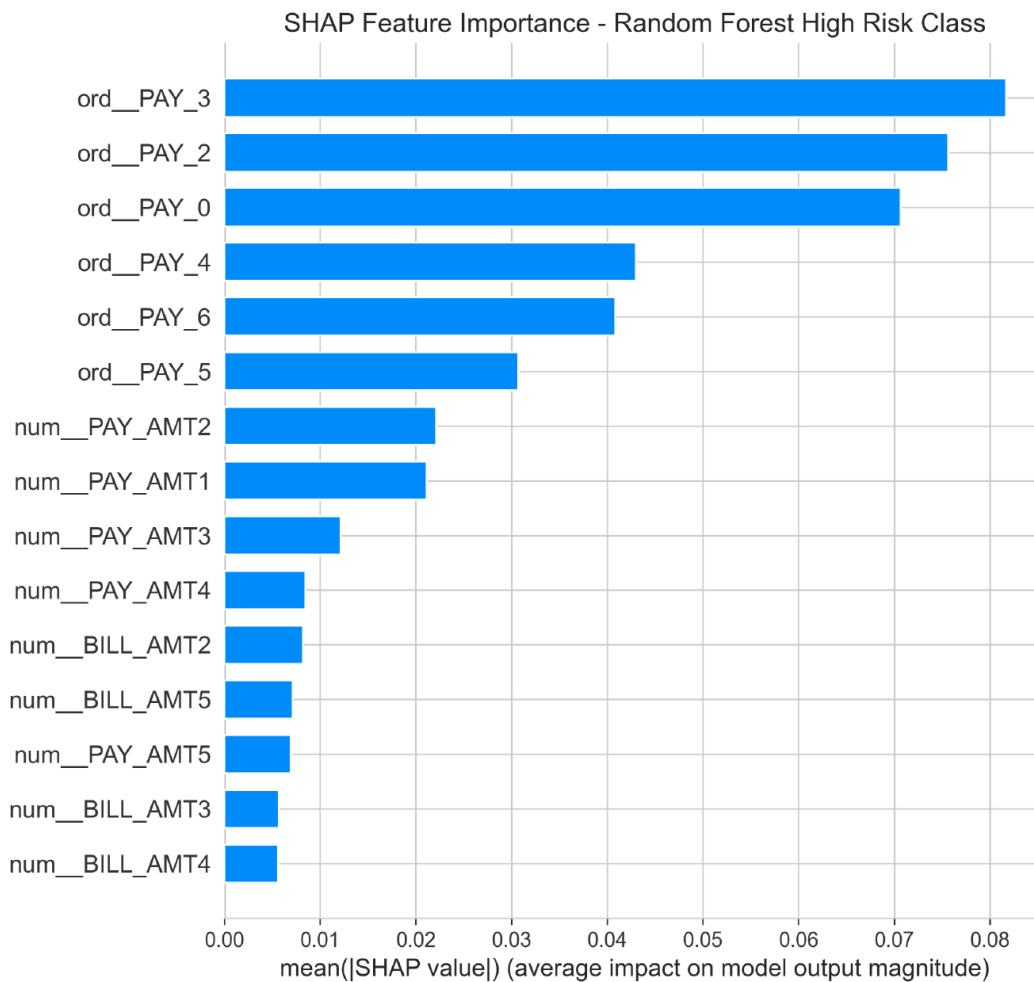


Figure 40. Random Forest SHAP Feature Importance for the High Risk Cluster

Repayment-status variables dominate the Random Forest importance ranking, led by PAY_3, PAY_2, and PAY_0. PAY_AMT2 and PAY_AMT1 are the most important payment-amount variables, while demographic features have very small average effects. This supports a behavior-centered interpretation of the High Risk cluster.

4.35 Observed Default Rate by Logistic Risk Group

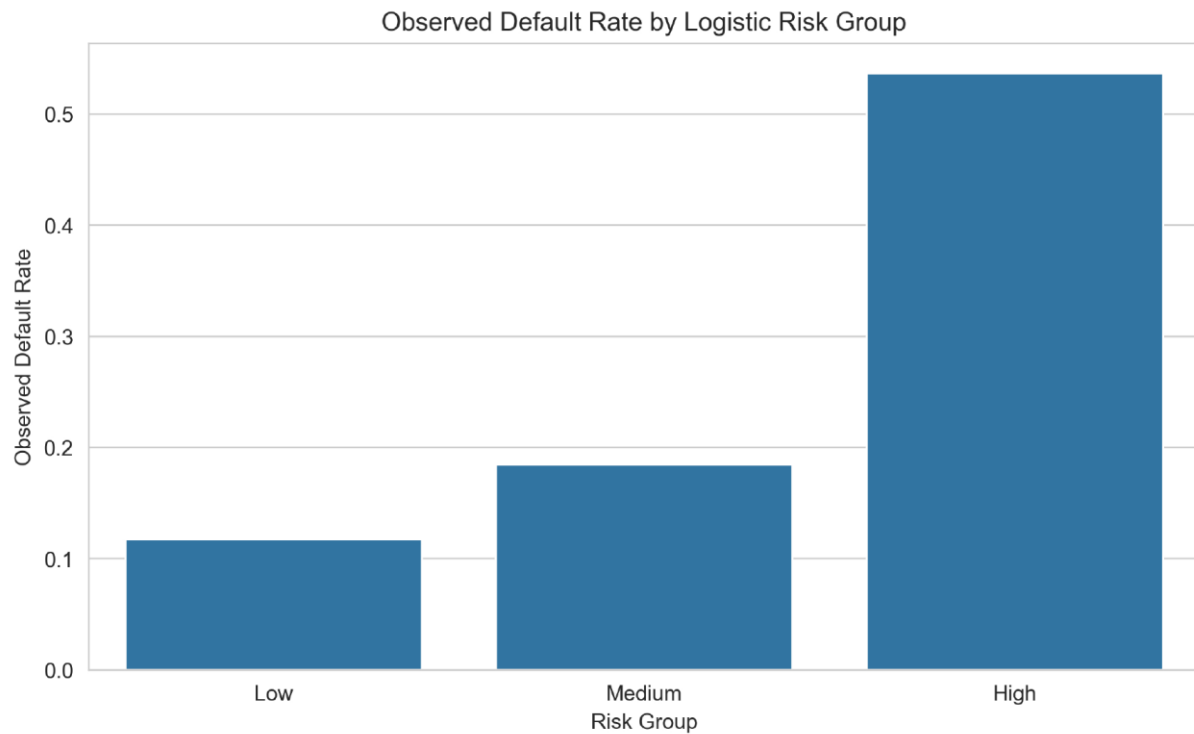


Figure 41. Observed Default Rate by Logistic Risk Group

Observed default rates rise clearly from 11.71% in the Low Risk group to 18.42% in Medium Risk and 53.67% in High Risk. This monotonic increase shows that the validation-based probability boundaries create useful risk ordering on the untouched test set. The groups can therefore support prioritization, subject to calibration, fairness checks, and human review.